

- 9.22** Consider a horizontal plane-parallel slab of gas of thickness L that is maintained at a constant temperature T . Assume that the gas has optical depth $\tau_{\lambda,0}$, with $\tau_{\lambda} = 0$ at the top surface of the slab. Assume further that incident radiation of intensity $I_{\lambda,0}$ enters the bottom of the slab from outside. Use the general solution of the transfer equation (9.54) to show that when looking at the slab from above, you see blackbody radiation if $\tau_{\lambda,0} \gg 1$. If $\tau_{\lambda,0} \ll 1$, show that you see absorption lines superimposed on the spectrum of the incident radiation if $I_{\lambda,0} > S_{\lambda}$ and emission lines superimposed on the spectrum of the incident radiation if $I_{\lambda,0} < S_{\lambda}$. (These latter two cases correspond to the spectral lines formed in the Sun's photosphere and chromosphere, respectively; see Section 11.2.) You may assume that the source function, S_{λ} , does not vary with position inside the gas. You may also assume thermodynamic equilibrium when $\tau_{\lambda,0} \gg 1$.
- 9.23** Verify that if the source function is $S_{\lambda} = a_{\lambda} + b_{\lambda} \tau_{\lambda,v}$, then the emergent intensity is given by Eq. (9.57), $I_{\lambda}(0) = a_{\lambda} + b_{\lambda} \cos \theta$.
- 9.24** Suppose that the shape of a spectral line is fit with one-half of an ellipse, such that the semimajor axis a is equal to the maximum depth of the line (let $F_{\lambda} = 0$) and the minor axis $2b$ is equal to the maximum width of the line (where it joins the continuum). What is the equivalent width of this line? *Hint:* You may find Eq. (2.4) useful.
- 9.25** Derive Eq. (9.60) for the uncertainty in the wavelength of a spectral line due to Heisenberg's uncertainty principle.
- 9.26** The two solar absorption lines given in Table 9.3 are produced when an electron makes an upward transition from the ground state orbital of the neutral Na I atom.
- Using the general curve of growth for the Sun, Fig. 9.22, repeat the procedure of Example 9.5.5 to find N_a , the number of absorbing sodium atoms per unit area of the photosphere.
 - Combine your results with those of Example 9.5.5 to find an average value of N_a . Use this value to plot the positions of the four sodium absorption lines on Fig. 9.22, and confirm that they do all lie on the curve of growth.
- 9.27** Pressure broadening (due to the presence of the electric fields of nearby ions) is unusually effective for the spectral lines of hydrogen. Using the general curve of growth for the Sun with these broad hydrogen absorption lines will result in an overestimate of the amount of hydrogen present. The following calculation nevertheless demonstrates just how abundant hydrogen is in the Sun.
- The two solar absorption lines given in Table 9.4 belong to the Paschen series, produced when an electron makes an upward transition from the $n = 3$ orbital of the hydrogen atom.
- Using the general curve of growth for the Sun, Fig. 9.22, repeat the procedure of Example 9.5.5 to find N_a , the number of absorbing hydrogen atoms per unit area of the photosphere (those with electrons initially in the $n = 3$ orbital).
 - Use the Boltzmann and Saha equations to calculate the total number of hydrogen atoms above each square meter of the Sun's photosphere.

TABLE 9.3 Data for Solar Sodium Lines for Problem 9.26. (Data from Aller, *Atoms, Stars, and Nebulae*, Revised Edition, Harvard University Press, Cambridge, MA, 1971.)

λ (nm)	W (nm)	f
330.298	0.0067	0.0049
589.594	0.0560	0.325

TABLE 9.4 Data for Solar Hydrogen Lines for Problem 9.27. (Data from Aller, *Atoms, Stars, and Nebulae*, Revised Edition, Harvard University Press, Cambridge, MA, 1971.)

λ (nm)	W (nm)	f
1093.8 (Pa γ)	0.22	0.0554
1004.9 (Pa δ)	0.16	0.0269

COMPUTER PROBLEMS

9.28 In this problem, you will use the values of the density and opacity at various points near the surface of the star to calculate the optical depth of these points. The data in Table 9.5 were obtained from the stellar model building program *StatStar*, described in Section 10.5 and Appendix L. The first point listed is at the surface of the stellar model.

- (a) Find the optical depth at each point by numerically integrating Eq. (9.15). Use a simple trapezoidal rule such that

$$d\tau = -\kappa\rho ds$$

becomes

$$\tau_{i+1} - \tau_i = -\left(\frac{\kappa_i \rho_i + \kappa_{i+1} \rho_{i+1}}{2}\right)(r_{i+1} - r_i),$$

where i and $i + 1$ designate adjacent zones in the model. Note that because s is measured along the path traveled by the photons, $ds = dr$.

- (b) Make a graph of the temperature (vertical axis) vs. the optical depth (horizontal axis).
(c) For each value of the optical depth, use Eq. (9.53) to calculate the temperature for a plane-parallel gray atmosphere in LTE. Plot these values of T on the same graph.
(d) The *StatStar* program utilizes a simplifying assumption that the surface temperature is zero (see Appendix L). Comment on the validity of the surface value of T that you found.
- 9.29** The binary star code *TwoStars*, discussed in Section 7.3 and Appendix K, makes use of an empirical limb darkening formula developed by W. Van Hamme (*Astronomical Journal*, 106, 1096, 1993):

$$\frac{I(\theta)}{I(\theta = 0)} = 1 - x(1 - \cos \theta) - y \cos \theta \log_{10}(\cos \theta),$$

where $x = 0.648$ and $y = 0.207$ for solar-type stars (other coefficients are provided for other types of stars).

- (a) Plot Van Hamme's formula for limb darkening over the range $0 \leq \theta \leq 90^\circ$. (Be sure to correctly treat the singularity in the function at $\theta = 90^\circ$.)
(b) Plot Eq. (9.58), which is based on the Eddington approximation, on the same graph.
(c) Where is the difference between the two formulae the greatest?
(d) Compare the two curves to the observational data shown in Fig. 9.17. Which curve best represents the solar data?

TABLE 9.5 A $1 M_{\odot}$ StatStar Model for Problem 9.28. $T_e = 5504$ K.

i	r (m)	T (K)	ρ (kg m $^{-3}$)	κ (m 2 kg $^{-1}$)
0	7.100764E+08	0.000000E+00	0.000000E+00	0.000000E+00
1	7.093244E+08	3.379636E+03	2.163524E-08	2.480119E+01
2	7.092541E+08	3.573309E+03	3.028525E-08	2.672381E+01
3	7.091783E+08	3.826212E+03	4.206871E-08	2.737703E+01
4	7.090959E+08	4.133144E+03	5.814973E-08	2.708765E+01
5	7.090062E+08	4.488020E+03	8.015188E-08	2.625565E+01
6	7.089085E+08	4.887027E+03	1.103146E-07	2.517004E+01
7	7.088019E+08	5.329075E+03	1.517126E-07	2.399474E+01
8	7.086856E+08	5.815187E+03	2.085648E-07	2.281158E+01
9	7.085588E+08	6.347784E+03	2.866621E-07	2.165611E+01
10	7.084205E+08	6.930293E+03	3.939580E-07	2.054686E+01
11	7.082697E+08	7.566856E+03	5.413734E-07	1.948823E+01
12	7.081052E+08	8.262201E+03	7.439096E-07	1.848131E+01
13	7.079259E+08	9.021603E+03	1.022171E-06	1.752513E+01
14	7.077303E+08	9.850881E+03	1.404459E-06	1.661785E+01
15	7.075169E+08	1.075642E+04	1.929644E-06	1.575731E+01
16	7.072843E+08	1.174520E+04	2.651111E-06	1.494128E+01
17	7.070306E+08	1.282486E+04	3.642174E-06	1.416754E+01
18	7.067540E+08	1.400375E+04	5.003513E-06	1.343396E+01
19	7.064524E+08	1.529096E+04	6.873380E-06	1.273849E+01
20	7.061235E+08	1.669643E+04	9.441600E-06	1.207917E+01
21	7.057649E+08	1.823102E+04	1.296880E-05	1.145414E+01
22	7.053741E+08	1.990656E+04	1.781279E-05	1.086165E+01
23	7.049480E+08	2.173599E+04	2.446473E-05	1.030001E+01
24	7.044836E+08	2.373341E+04	3.359882E-05	9.767631E+00
25	7.039774E+08	2.591421E+04	4.614038E-05	9.263005E+00
26	7.034259E+08	2.829519E+04	6.335925E-05	8.784696E+00
27	7.028250E+08	3.089468E+04	8.699788E-05	8.331344E+00
28	7.021704E+08	3.373266E+04	1.194469E-04	7.901659E+00
29	7.014574E+08	3.683096E+04	1.639859E-04	7.494416E+00
30	7.006810E+08	4.021337E+04	2.251132E-04	7.108452E+00
31	6.998356E+08	4.390583E+04	3.089976E-04	6.742665E+00
32	6.989155E+08	4.793666E+04	4.240980E-04	6.396010E+00
33	6.979141E+08	5.233670E+04	5.820105E-04	6.067495E+00
34	6.968247E+08	5.713961E+04	7.986295E-04	5.756179E+00
35	6.956399E+08	6.238205E+04	1.095736E-03	5.461170E+00
36	6.943518E+08	6.810401E+04	1.503169E-03	5.181621E+00
37	6.929517E+08	7.434904E+04	2.061803E-03	4.916730E+00
38	6.914307E+08	8.116461E+04	2.827602E-03	4.665735E+00
39	6.897790E+08	8.860239E+04	3.877181E-03	4.427914E+00
40	6.879861E+08	9.671869E+04	5.315384E-03	4.202584E+00
41	6.860411E+08	1.055748E+05	7.285639E-03	3.989094E+00

The Interiors of Stars

- 10.1 *Hydrostatic Equilibrium*
- 10.2 *Pressure Equation of State*
- 10.3 *Stellar Energy Sources*
- 10.4 *Energy Transport and Thermodynamics*
- 10.5 *Stellar Model Building*
- 10.6 *The Main Sequence*

10.1 ■ HYDROSTATIC EQUILIBRIUM

In the last two chapters, many of the observational details of stellar spectra were discussed, along with the basic physical principles behind the production of the observed lines. Analysis of that light, collected by ground-based and space-based telescopes, enables astronomers to determine a variety of quantities related to the outer layers of stars, such as effective temperature, luminosity, and composition. However, with the exceptions of the ongoing detection of neutrinos from the Sun (which will be discussed later in this chapter and in Chapter 11) and the one-time detection from Supernova 1987A (Section 15.3), no direct way exists to observe the central regions of stars.

Determining the Internal Structures of Stars

To deduce the detailed internal structure of stars requires the generation of computer models that are consistent with all known physical laws and that ultimately agree with observable surface features. Although much of the theoretical foundation of stellar structure was understood by the first half of the twentieth century, it wasn't until the 1960s that sufficiently fast computing machines became available to carry out all of the necessary calculations. Arguably one of the greatest successes of theoretical astrophysics has been the detailed computer modeling of stellar structure and evolution. However, despite all of the successes of such calculations, numerous questions remain unanswered. The solution to many of these problems requires a more detailed theoretical understanding of the physical processes in operation in the interiors of stars, combined with even greater computational power.

The theoretical study of stellar structure, coupled with observational data, clearly shows that stars are dynamic objects, usually changing at an imperceptibly slow rate by human standards, although they can sometimes change in very rapid and dramatic ways, such as during a supernova explosion. That such changes must occur can be seen by simply considering the observed energy output of a star. In the Sun, 3.839×10^{26} J of energy is

emitted every second. This rate of energy output would be sufficient to melt a 0°C block of ice measuring $1 \text{ AU} \times 1 \text{ mile} \times 1 \text{ mile}$ in only 0.3 s , assuming that the absorption of the energy was 100% efficient. Because stars do not have infinite supplies of energy, they must eventually use up their reserves and die. *Stellar evolution is the result of a constant fight against the relentless pull of gravity.*

The Derivation of the Hydrostatic Equilibrium Equation

The gravitational force is always attractive, implying that an opposing force must exist if a star is to avoid collapse. This force is provided by pressure. To calculate how the pressure must vary with depth, consider a cylinder of mass dm whose base is located a distance r from the center of a spherical star (see Fig. 10.1). The areas of the top and bottom of the cylinder are each A and the cylinder's height is dr . Furthermore, assume that the only forces acting on the cylinder are gravity and the pressure force, which is always normal to the surface and may vary with distance from the center of the star. Using Newton's second law $\mathbf{F} = m\mathbf{a}$, we have the net force on the cylinder:

$$dm \frac{d^2r}{dt^2} = F_g + F_{P,t} + F_{P,b},$$

where $F_g < 0$ is the gravitational force directed inward and $F_{P,t}$ and $F_{P,b}$ are the pressure forces on the top and bottom of the cylinder, respectively. Note that since the pressure forces on the side of the cylinder will cancel, they have been explicitly excluded from the expression. Because the pressure force is always normal to the surface, the force exerted on the top of the cylinder must necessarily be directed toward the center of the star ($F_{P,t} < 0$),

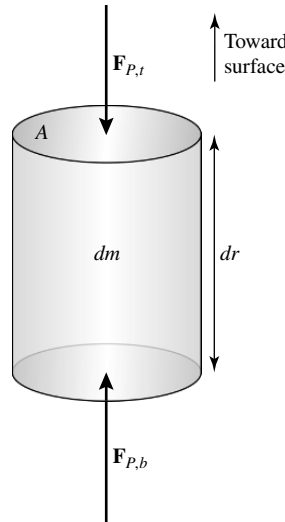


FIGURE 10.1 In a static star the gravitational force on a mass element is exactly canceled by the outward force due to a pressure gradient in the star. A cylinder of mass dm is located at a distance r from the center of the star. The height of the cylinder is dr , and the areas of the top and bottom are both A . The density of the gas is assumed to be ρ at that position.

whereas the force on the bottom is directed outward ($F_{P,b} > 0$). Writing $F_{P,t}$ in terms of $F_{P,b}$ and a correction term dF_P that accounts for the change in force due to a change in r results in

$$F_{P,t} = -(F_{P,b} + dF_P).$$

Substitution into the previous expression gives

$$dm \frac{d^2 r}{dt^2} = F_g - dF_P. \quad (10.1)$$

As we noted in Example 2.2.1, the gravitational force on a small mass dm located at a distance r from the center of a spherically symmetric mass is

$$F_g = -G \frac{M_r dm}{r^2}, \quad (10.2)$$

where M_r is the mass inside the sphere of radius r , often referred to as the *interior mass*. The contribution to the gravitational force by spherically symmetric mass shells located outside r is zero (the proof of this is left to Problem 10.2).

Pressure is defined as the amount of force per unit area exerted on a surface, or

$$P \equiv \frac{F}{A}.$$

Allowing for a difference in pressures dP between the top of the cylinder and the bottom due to the different forces exerted on each surface, the differential force may be expressed as

$$dF_P = A dP. \quad (10.3)$$

Substituting Eqs. (10.2) and (10.3) into Eq. (10.1) gives

$$dm \frac{d^2 r}{dt^2} = -G \frac{M_r dm}{r^2} - A dP. \quad (10.4)$$

If the density of the gas in the cylinder is ρ , its mass is just

$$dm = \rho A dr,$$

where $A dr$ is the cylinder's volume. Using this expression in Eq. (10.4) yields

$$\rho A dr \frac{d^2 r}{dt^2} = -G \frac{M_r \rho A dr}{r^2} - A dP.$$

Finally, dividing through by the volume of the cylinder, we have

$$\rho \frac{d^2 r}{dt^2} = -G \frac{M_r \rho}{r^2} - \frac{dP}{dr}. \quad (10.5)$$

This is the equation for the radial motion of the cylinder, assuming spherical symmetry.

If we assume further that the star is static, then the acceleration must be zero. In this case Eq. (10.5) reduces to

$$\boxed{\frac{dP}{dr} = -G \frac{M_r \rho}{r^2} = -\rho g,} \quad (10.6)$$

where $g \equiv GM_r/r^2$ is the local acceleration of gravity at radius r . Equation (10.6), the condition of **hydrostatic equilibrium**, represents one of the fundamental equations of stellar structure for spherically symmetric objects under the assumption that accelerations are negligible. Equation (10.6) clearly indicates that in order for a star to be static, a *pressure gradient* dP/dr must exist to counteract the force of gravity. It is not the pressure that supports a star, but the change in pressure with radius. Furthermore, the pressure must *decrease* with increasing radius; the pressure is necessarily larger in the interior than it is near the surface.

Example 10.1.1. To obtain a very crude estimate of the pressure at the center of the Sun, assume that $M_r = 1 M_\odot$, $r = 1 R_\odot$, and $\rho = \bar{\rho}_\odot = 1410 \text{ kg m}^{-3}$ is the *average* solar density (see Example 8.2.1). Assume also that the surface pressure is exactly zero. Then, converting the differential equation to a difference equation, the left hand side of Eq. (10.6) becomes

$$\frac{dP}{dr} \sim \frac{P_s - P_c}{R_s - 0} \sim -\frac{P_c}{R_\odot},$$

where P_c is the central pressure, and P_s and R_s are the surface pressure and radius, respectively. Substituting into the equation of hydrostatic equilibrium and solving for the central pressure, we find

$$P_c \sim G \frac{M_\odot \bar{\rho}_\odot}{R_\odot} \sim 2.7 \times 10^{14} \text{ N m}^{-2}.$$

To obtain a more accurate value, we need to *integrate* the hydrostatic equilibrium equation from the surface to the center, taking into consideration the change in the interior mass M_r at each point, together with the variation of density with radius $\rho_r \equiv \rho(r)$, giving

$$\int_{P_s}^{P_c} dP = P_c = - \int_{R_s}^{R_c} \frac{GM_r \rho}{r^2} dr.$$

Actually carrying out the integration requires functional forms of M_r and ρ . Unfortunately, such explicit expressions are not available, implying that further relationships between such quantities must be developed.

From a more rigorous calculation, a solar model gives a central pressure of nearly $2.34 \times 10^{16} \text{ N m}^{-2}$. This value is much larger than the one obtained from our crude estimate because of the increased density near the center of the Sun. As a reference, one atmosphere of pressure is $1 \text{ atm} = 1.013 \times 10^5 \text{ N m}^{-2}$; therefore, the more realistic model predicts a central pressure of $2.3 \times 10^{11} \text{ atm}$!

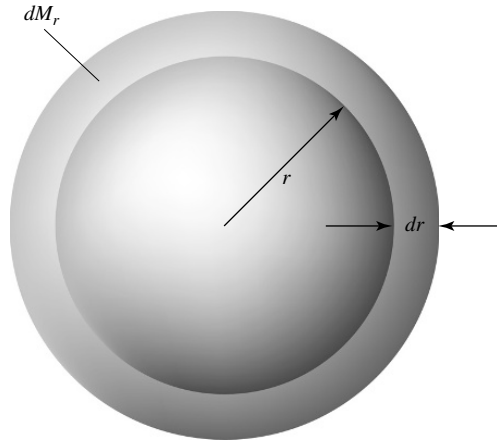


FIGURE 10.2 A spherically symmetric shell of mass dM_r having a thickness dr and located a distance r from the center of the star. The local density of the shell is ρ .

The Equation of Mass Conservation

A second relationship involving mass, radius, and density also exists. Again, for a spherically symmetric star, consider a *shell* of mass dM_r and thickness dr , located a distance r from the center, as in Fig. 10.2. Assuming that the shell is sufficiently thin (i.e., $dr \ll r$), the volume of the shell is approximately $dV = 4\pi r^2 dr$. If the local density of the gas is ρ , the shell's mass is given by

$$dM_r = \rho(4\pi r^2 dr).$$

Rewriting, we arrive at the **mass conservation equation**,

$$\boxed{\frac{dM_r}{dr} = 4\pi r^2 \rho}, \quad (10.7)$$

which dictates how the interior mass of a star must change with distance from the center. Equation (10.7) is the second of the fundamental equations of stellar structure.

10.2 ■ PRESSURE EQUATION OF STATE

Up to this point no information has been provided about the origin of the pressure term required by Eq. (10.6). To describe this macroscopic manifestation of particle interactions, it is necessary to derive a pressure **equation of state** of the material. Such an equation of state relates the dependence of pressure on other fundamental parameters of the material. One well-known example of a pressure equation of state is the **ideal gas law**, often expressed as

$$PV = NkT,$$

where V is the volume of the gas, N is the number of particles, T is the temperature, and k is Boltzmann's constant.

Although this expression was first determined experimentally, it is informative to derive it from fundamental physical principles. The approach used here will also provide a general method for considering environments where the assumptions of the ideal gas law do not apply, a situation frequently encountered in astrophysical problems.

The Derivation of the Pressure Integral

Consider a cylinder of gas of length Δx and cross-sectional area A , as in Fig. 10.3. The gas contained in the cylinder is assumed to be composed of point particles, each of mass m , that interact through perfectly elastic collisions only—in other words, as an ideal gas. To determine the pressure exerted on one of the ends of the container, examine the result of an impact on the right wall by an individual particle. Since, for a perfectly elastic collision, the angle of reflection from the wall must be equal to the angle of incidence, the change in momentum of the particle is necessarily entirely in the x -direction, normal to the surface. From Newton's second law¹ ($\mathbf{f} = m\mathbf{a} = d\mathbf{p}/dt$) and third law, the *impulse* $\mathbf{f} \Delta t$ delivered to the wall is just the negative of the change in momentum of the particle, or

$$\mathbf{f} \Delta t = -\Delta \mathbf{p} = 2p_x \hat{\mathbf{i}},$$

where p_x is the component of the particle's initial momentum in the x -direction. Now the average force exerted by the particle over a period of time can be determined by evaluating the time interval between collisions with the right wall. Since the particle must traverse the length of the container twice before returning for a second reflection, the time interval between collisions with the same wall by the same particle is given by

$$\Delta t = 2 \frac{\Delta x}{v_x},$$

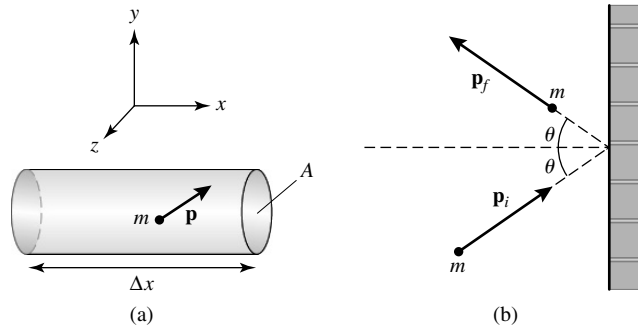


FIGURE 10.3 (a) A cylinder of gas of length Δx and cross-sectional area A . Assume that the gas contained in the cylinder is an ideal gas. (b) The collision of an individual point mass with one of the ends of the cylinder. For a perfectly elastic collision, the angle of reflection must equal the angle of incidence.

¹Note that a lowercase $\mathbf{f}$ is used here to indicate that the force is due to a *single* particle.

so that the average force exerted on the wall by a single particle over that time period is given by

$$f = \frac{2p_x}{\Delta t} = \frac{p_x v_x}{\Delta x},$$

where it is assumed that the direction of the force vector is normal to the surface.

Now, because $p_x \propto v_x$, the numerator is proportional to v_x^2 . To evaluate this, recall that the magnitude of the velocity vector is given by $v^2 = v_x^2 + v_y^2 + v_z^2$. For a sufficiently large collection of particles in random motion, the likelihood of motion in each of the three directions is the same, or $\overline{v_x^2} = \overline{v_y^2} = \overline{v_z^2} = v^2/3$. Substituting $\frac{1}{3}pv$ for $p_x v_x$, the average force per particle having momentum p is

$$f(p) = \frac{1}{3} \frac{pv}{\Delta x}.$$

It is usually the case that the particles have a range of momenta. If the number of particles with momenta between p and $p + dp$ is given by the expression $N_p dp$, then the total number of particles in the container is

$$N = \int_0^\infty N_p dp.$$

The contribution to the total force, $dF(p)$, by *all* particles in that momentum range is given by

$$dF(p) = f(p)N_p dp = \frac{1}{3} \frac{N_p}{\Delta x} pv dp.$$

Integrating over all possible values of the momentum, the total force exerted by particle collisions is

$$F = \frac{1}{3} \int_0^\infty \frac{N_p}{\Delta x} pv dp.$$

Dividing both sides of the expression by the surface area of the wall A gives the pressure on the surface as $P = F/A$. Noting that $\Delta V = A \Delta x$ is just the volume of the cylinder and defining $n_p dp$ to be the number of particles *per unit volume* having momenta between p and $p + dp$, or

$$n_p dp \equiv \frac{N_p}{\Delta V} dp,$$

we find that the pressure exerted on the wall is

$$P = \frac{1}{3} \int_0^\infty n_p pv dp. \quad (10.8)$$

This expression, which is sometimes called the **pressure integral**, makes it possible to compute the pressure, given some *distribution function*, $n_p dp$.

The Ideal Gas Law in Terms of the Mean Molecular Weight

Equation (10.8) is valid for both massive and massless particles (such as photons) traveling at any speed. For the special case of massive, nonrelativistic particles, we may use $p = mv$ to write the pressure integral as

$$P = \frac{1}{3} \int_0^\infty m n_v v^2 dv, \quad (10.9)$$

where $n_v dv = n_p dp$ is the number of particles per unit volume having speeds between v and $v + dv$.

The function $n_v dv$ is dependent on the physical nature of the system being described. In the case of an ideal gas, $n_v dv$ is the Maxwell–Boltzmann velocity distribution described in Chapter 8 (Eq. 8.1),

$$n_v dv = n \left(\frac{m}{2\pi kT} \right)^{3/2} e^{-mv^2/2kT} 4\pi v^2 dv,$$

where $n = \int_0^\infty n_v dv$ is the particle number density. Substituting into the pressure integral finally gives

$$P_g = nkT \quad (10.10)$$

(the proof is left as an exercise in Problem 10.5). Since $n \equiv N/V$, Eq. (10.10) is just the familiar ideal gas law.

In astrophysical applications it is often convenient to express the ideal gas law in an alternative form. Since n is the particle number density, it is clear that it must be related to the mass density of the gas. Allowing for a variety of particles of different masses, it is then possible to express n as

$$n = \frac{\rho}{\bar{m}},$$

where $\bar{m}$ is the average mass of a gas particle. Substituting, the ideal gas law becomes

$$P_g = \frac{\rho kT}{\bar{m}}.$$

We now define a new quantity, the **mean molecular weight**, as

$$\mu \equiv \frac{\bar{m}}{m_H},$$

where $m_H = 1.673532499 \times 10^{-27}$ kg is the mass of the hydrogen atom. *The mean molecular weight is just the average mass of a free particle in the gas, in units of the mass of hydrogen.* The ideal gas law can now be written in terms of the mean molecular weight as

$$\boxed{P_g = \frac{\rho kT}{\mu m_H}}. \quad (10.11)$$

The mean molecular weight depends on the composition of the gas as well as on the state of ionization of each species. The level of ionization enters because free electrons must be included in the average mass per particle $\bar{m}$. This implies that a detailed analysis of the Saha equation (8.8) is necessary to calculate the relative numbers of ionization states. When the gas is either completely neutral or completely ionized, the calculation simplifies significantly, however.

For a completely neutral gas,

$$\bar{m}_n = \frac{\sum_j N_j m_j}{\sum_j N_j}, \quad (10.12)$$

where m_j and N_j are, respectively, the mass and the total number of atoms of type j that are present in the gas, and the sums are assumed to be carried out over all types of atoms. Dividing by m_H yields

$$\mu_n = \frac{\sum_j N_j A_j}{\sum_j N_j},$$

where $A_j \equiv m_j/m_H$. Similarly, for a completely ionized gas,

$$\mu_i \simeq \frac{\sum_j N_j A_j}{\sum_j N_j (1 + z_j)},$$

where $1 + z_j$ accounts for the nucleus plus the number of free electrons that result from completely ionizing an atom of type j . (Do not confuse z_j with Z , the mass fraction of metals.)

By inverting the expression for $\bar{m}$, it is possible to write alternative equations for μ in terms of mass fractions.² Recalling that $\bar{m} = \mu m_H$, Eq. (10.12) for a neutral gas gives

$$\begin{aligned} \frac{1}{\mu_n m_H} &= \frac{\sum_j N_j}{\sum_j N_j m_j} \\ &= \frac{\text{total number of particles}}{\text{total mass of gas}} \end{aligned}$$

²Recall the definitions of mass fractions on page 250.

$$\begin{aligned}
&= \sum_j \frac{\text{number of particles from } j}{\text{mass of particles from } j} \cdot \frac{\text{mass of particles from } j}{\text{total mass of gas}} \\
&= \sum_j \frac{N_j}{N_j A_j m_H} X_j \\
&= \sum_j \frac{1}{A_j m_H} X_j,
\end{aligned}$$

where X_j is the mass fraction of atoms of type j . Solving for $1/\mu_n$, we have

$$\frac{1}{\mu_n} = \sum_j \frac{1}{A_j} X_j. \quad (10.13)$$

Thus, for a neutral gas,

$$\frac{1}{\mu_n} \simeq X + \frac{1}{4}Y + \left\langle \frac{1}{A} \right\rangle_n Z. \quad (10.14)$$

$\langle 1/A \rangle_n$ is a weighted average of all elements in the gas heavier than helium. For solar abundances, $\langle 1/A \rangle_n \sim 1/15.5$.

The mean molecular weight of a completely ionized gas may be determined in a similar way. It is necessary only to include the *total* number of particles contained in the sample, both nuclei and electrons. For instance, each hydrogen atom contributes one free electron, together with its nucleus, to the total number of particles. Similarly, one helium atom contributes two free electrons plus its nucleus. Therefore, for a completely ionized gas, Eq. (10.13) becomes

$$\frac{1}{\mu_i} = \sum_j \frac{1 + z_j}{A_j} X_j. \quad (10.15)$$

Including hydrogen and helium explicitly, we have

$$\frac{1}{\mu_i} \simeq 2X + \frac{3}{4}Y + \left\langle \frac{1+z}{A} \right\rangle_i Z. \quad (10.16)$$

For elements much heavier than helium, $1 + z_j \simeq z_j$, where $z_j \gg 1$ represents the number of protons (or electrons) in an atom of type j . It also holds that $A_j \simeq 2z_j$, the relation being based on the facts that sufficiently massive atoms have approximately the same number of protons and neutrons in their nuclei and that protons and neutrons have very similar masses (see page 299). Thus

$$\left\langle \frac{1+z}{A} \right\rangle_i \simeq \frac{1}{2}.$$

If we assume that $X = 0.70$, $Y = 0.28$, and $Z = 0.02$, a composition typical of younger stars, then with these expressions for the mean molecular weight, $\mu_n = 1.30$ and $\mu_i = 0.62$.

The Average Kinetic Energy Per Particle

Further investigation of the ideal gas law shows that it is also possible to combine Eq. (10.10) with the pressure integral (Eq. 10.9) to find the average kinetic energy per particle. Equating, we see that

$$nkT = \frac{1}{3} \int_0^\infty mn_v v^2 dv.$$

This expression can be rewritten to give

$$\frac{1}{n} \int_0^\infty n_v v^2 dv = \frac{3kT}{m}.$$

However, the left-hand side of this expression is just the integral average of v^2 weighted by the Maxwell–Boltzmann distribution function. Thus

$$\overline{v^2} = \frac{3kT}{m},$$

or

$$\frac{1}{2} m \overline{v^2} = \frac{3}{2} kT. \quad (10.17)$$

It is worth noting that the factor of 3 arose from averaging particle velocities over the three coordinate directions (or *degrees of freedom*), which was performed on page 290. Thus the average kinetic energy of a particle is $\frac{1}{2}kT$ per degree of freedom.

Fermi–Dirac and Bose–Einstein Statistics

As has already been mentioned, there are stellar environments where the assumptions of the ideal gas law do not hold even approximately. For instance, in the pressure integral it was assumed that the upper limit of integration for velocity was infinity. Of course, this cannot be the case since, from Einstein’s theory of special relativity, the maximum possible value of velocity is c , the speed of light. Furthermore, the effects of quantum mechanics were also neglected in the derivation of the ideal gas law. When the Heisenberg uncertainty principle and the Pauli exclusion principle are considered, a distribution function different from the Maxwell–Boltzmann distribution results. The **Fermi–Dirac** distribution function considers these important principles and leads to a very different pressure equation of state when applied to extremely dense matter such as that found in white dwarf stars and neutron stars. These exotic objects will be discussed in detail in Chapter 16. As mentioned in Section 5.4, particles such as electrons, protons, and neutrons that obey **Fermi–Dirac statistics** are called **fermions**.

Another statistical distribution function is obtained if it is assumed that the presence of some particles in a particular state enhances the likelihood of others being in the same state, an effect somewhat opposite to that of the Pauli exclusion principle. **Bose–Einstein statistics** has a variety of applications, including understanding the behavior of photons. Particles that obey Bose–Einstein statistics are known as **bosons**.

Just as special relativity and quantum mechanics must give classical results in the appropriate limits, Fermi–Dirac and Bose–Einstein statistics also approach the classical regime at

sufficiently low densities and velocities. In these limits both distribution functions become indistinguishable from the classical Maxwell–Boltzmann distribution function.

The Contribution Due to Radiation Pressure

Because photons possess momentum $p_\gamma = h\nu/c$ (Eq. 5.5), they are capable of delivering an impulse to other particles during absorption or reflection. Consequently, electromagnetic radiation results in another form of pressure. It is instructive to rederive the expression for radiation pressure found in Chapter 9 by making use of the pressure integral. Substituting the speed of light for the velocity v , using the expression for photon momentum, and using an identity for the distribution function, $n_p dp = n_\nu d\nu$, the general pressure integral, Eq. (10.8), now describes the effect of radiation, giving

$$P_{\text{rad}} = \frac{1}{3} \int_0^\infty h\nu n_\nu d\nu.$$

At this point, the problem again reduces to finding an appropriate expression for $n_\nu d\nu$. Since photons are bosons, the Bose–Einstein distribution function would apply. However, the problem may also be solved by realizing that $n_\nu d\nu$ represents the number density of photons having frequencies lying in the range between ν and $\nu + d\nu$. Multiplying by the energy of each photon in that range would then give the *energy density* over the frequency interval, or

$$P_{\text{rad}} = \frac{1}{3} \int_0^\infty u_\nu d\nu, \quad (10.18)$$

where $u_\nu d\nu = h\nu n_\nu d\nu$. But the energy density distribution function is found from the Planck function for blackbody radiation, Eq. (9.6). Substituting into Eq. (10.18) and performing the integration lead to

$$P_{\text{rad}} = \frac{1}{3} a T^4, \quad (10.19)$$

where a is the radiation constant found previously in Eq. (9.7).

In many astrophysical situations the pressure due to photons can actually exceed by a significant amount the pressure produced by the gas. In fact it is possible that the magnitude of the force due to radiation pressure can become sufficiently great that it surpasses the gravitational force, resulting in an overall expansion of the system.

Combining both the ideal gas and radiation pressure terms, the total pressure becomes

$$P_t = \frac{\rho k T}{\mu m_H} + \frac{1}{3} a T^4. \quad (10.20)$$

Example 10.2.1. Using the results of Example 10.1.1, we can estimate the central temperature of the Sun. Neglecting the radiation pressure term, the central temperature is found

continued

from the ideal gas law equation of state to be

$$T_c = \frac{P_c \mu m_H}{\rho k}.$$

Using $\bar{\rho}_\odot$, a value of $\mu_i = 0.62$ appropriate for complete ionization,³ and the estimated value for the central pressure, we find that

$$T_c \sim 1.44 \times 10^7 \text{ K}$$

which is in reasonable agreement with more detailed calculations. One solar model gives a central temperature of $1.57 \times 10^7 \text{ K}$. At this temperature, the pressure due to radiation is only $1.53 \times 10^{13} \text{ N m}^{-2}$, 0.065% of the gas pressure.

10.3 ■ STELLAR ENERGY SOURCES

As we have already seen, the rate of energy output of stars (their luminosities) is very large. However, the question of the source of that energy has not yet been addressed. Clearly, one measure of the lifetime of a star must be related to how long it can sustain its power output.

Gravitation and the Kelvin–Helmholtz Timescale

One likely source of stellar energy is gravitational potential energy. Recall that the gravitational potential energy of a system of two particles is given by Eq. (2.14),

$$U = -G \frac{Mm}{r}.$$

As the distance between M and m diminishes, the gravitational potential energy becomes *more negative*, implying that energy must have been converted to other forms, such as kinetic energy. If a star can manage to convert its gravitational potential energy into heat and then radiate that heat into space, the star may be able to shine for a significant period of time. However, we must also remember that by the virial theorem (Eq. 2.47) the total energy of a system of particles in equilibrium is one-half of the system's potential energy. Therefore, only one-half of the change in gravitational potential energy of a star is actually available to be radiated away; the remaining potential energy supplies the thermal energy that heats the star.

Calculating the gravitational potential energy of a star requires consideration of the interaction between every possible pair of particles. This is not as difficult as it might first seem. The gravitational force on a point mass dm_i located outside of a spherically symmetric mass M_r is

$$dF_{g,i} = G \frac{M_r dm_i}{r^2}$$

³Since, as we will see in the next chapter, the Sun has already converted a significant amount of its core hydrogen into helium via nuclear reactions, the actual value of μ_i is closer to 0.84.

and is directed toward the center of the sphere. This is just the same force that would exist if all of the mass of the sphere were located at its center, a distance r from the point mass. This immediately implies that the gravitational potential energy of the point mass is

$$dU_{g,i} = -G \frac{M_r dm_i}{r}.$$

If, rather than considering an individual point mass, we assume that point masses are distributed uniformly within a shell of thickness dr and mass dm (where dm is the sum of all the point masses dm_i), then

$$dm = 4\pi r^2 \rho dr,$$

where ρ is the mass density of the shell and $4\pi r^2 dr$ is its volume. Thus

$$dU_g = -G \frac{M_r 4\pi r^2 \rho}{r} dr.$$

Integrating over all mass shells from the center of the star to the surface, its total gravitational potential energy becomes

$$U_g = -4\pi G \int_0^R M_r \rho r dr, \quad (10.21)$$

where R is the radius of the star.

An exact calculation of U_g requires knowledge of how ρ , and consequently M_r , depend on r . Nevertheless, an approximate value can be obtained by assuming that ρ is constant and equal to its average value, or

$$\rho \sim \bar{\rho} = \frac{M}{\frac{4}{3}\pi R^3},$$

M being the total mass of the star. Now we may also approximate M_r as

$$M_r \sim \frac{4}{3}\pi r^3 \bar{\rho}.$$

If we substitute into Eq. (10.21), the total gravitational potential energy becomes

$$U_g \sim -\frac{16\pi^2}{15} G \bar{\rho}^2 R^5 \sim -\frac{3}{5} \frac{GM^2}{R}. \quad (10.22)$$

Lastly, applying the virial theorem, the total mechanical energy of the star is

$$E \sim -\frac{3}{10} \frac{GM^2}{R}. \quad (10.23)$$

Example 10.3.1. If the Sun were originally much larger than it is today, how much energy would have been liberated in its gravitational collapse? Assuming that its original radius

continued

was R_i , where $R_i \gg 1 R_\odot$, then the energy radiated away during collapse would be

$$\Delta E_g = -(E_f - E_i) \simeq -E_f \simeq \frac{3}{10} \frac{GM_\odot^2}{R_\odot} \simeq 1.1 \times 10^{41} \text{ J.}$$

Assuming also that the luminosity of the Sun has been roughly constant throughout its lifetime, it could emit energy at that rate for approximately

$$t_{\text{KH}} = \frac{\Delta E_g}{L_\odot} \quad (10.24)$$

$$\sim 10^7 \text{ yr.}$$

t_{KH} is known as the **Kelvin–Helmholtz timescale**. Based on radioactive dating techniques, however, the estimated age of rocks on the Moon's surface is over 4×10^9 yr. It seems unlikely that the age of the Sun is less than the age of the Moon! Therefore, gravitational potential energy alone cannot account for the Sun's luminosity throughout its entire lifetime. As we shall see in later chapters, however, gravitational energy can play an important role during some phases of the evolution of stars.

Another possible energy source involves chemical processes. However, since chemical reactions are based on the interactions of orbital electrons in atoms, the amount of energy available to be released per atom is not likely to be more than 1–10 electron volts, typical of the atomic energy levels in hydrogen and helium (see Section 5.3). Given the number of atoms present in a star, the amount of chemical energy available is also far too low to account for the Sun's luminosity over a reasonable period of time (Problem 10.3).

The Nuclear Timescale

The nuclei of atoms may also be considered as sources of energy. Whereas electron orbits involve energies in the electron volt (eV) range, nuclear processes generally involve energies millions of times larger (MeV). Just as chemical reactions can result in the transformation of atoms into molecules or one kind of molecule into another, nuclear reactions change one type of nucleus into another.

The nucleus of a particular **element** is specified by the number of protons, Z , it contains (not to be confused with the mass fraction of metals), with each proton carrying a charge of $+e$. Obviously, in a neutral atom the number of protons must exactly equal the number of orbital electrons. An **isotope** of a given element is identified by the number of neutrons, N , in the nucleus, with neutrons being electrically neutral, as the name implies. (All isotopes of a given element have the same number of protons.) Collectively, protons and neutrons are referred to as **nucleons**, the number of nucleons in a particular isotope being $A = Z + N$. Since protons and neutrons have very similar masses and greatly exceed the mass of electrons, A is a good indication of the mass of the isotope and is often referred to as the

mass number.⁴ The masses of the proton, neutron, and electron are, respectively,

$$m_p = 1.67262158 \times 10^{-27} \text{ kg} = 1.00727646688 \text{ u}$$

$$m_n = 1.67492716 \times 10^{-27} \text{ kg} = 1.00866491578 \text{ u}$$

$$m_e = 9.10938188 \times 10^{-31} \text{ kg} = 0.0005485799110 \text{ u}.$$

It is often convenient to express the masses of nuclei in terms of *atomic mass units*; $1 \text{ u} = 1.66053873 \times 10^{-27} \text{ kg}$, exactly one-twelfth the mass of the isotope carbon-12. The masses of nuclear particles are also frequently expressed in terms of their rest mass energies, in units of MeV. Using Einstein's $E = mc^2$, we find $1 \text{ u} = 931.494013 \text{ MeV}/c^2$. When masses are expressed simply in terms of rest mass energies, as is often the case, the factor c^2 is implicitly assumed.

The simplest isotope of hydrogen is composed of one proton and one electron and has a mass of $m_H = 1.00782503214 \text{ u}$. This mass is actually very slightly *less* than the combined masses of the proton and electron taken separately. In fact, if the atom is in its ground state, the exact mass difference is 13.6 eV, which is just its ionization potential. Since mass is equivalent to a corresponding amount of energy, and the total mass–energy of the system must be conserved, any loss in energy when the electron and proton combine to form an atom must come at the expense of a loss in total mass.

Similarly, energy is also released with an accompanying loss in mass when nucleons are combined to form atomic nuclei. A helium nucleus, composed of two protons and two neutrons, can be formed by a series of nuclear reactions originally involving four hydrogen nuclei (i.e., $4\text{H} \rightarrow \text{He} + \text{low mass remnants}$). Such reactions are known as **fusion** reactions, since lighter particles are “fused” together to form a heavier particle. (Conversely, a **fission** reaction occurs when a massive nucleus is split into smaller fragments.) The total mass of the four hydrogen atoms is 4.03130013 u , whereas the mass of one helium atom is $m_{\text{He}} = 4.002603 \text{ u}$. Neglecting the contribution of low-mass remnants such as neutrinos, the combined mass of the hydrogen atoms *exceeds* the mass of the helium atom by $\Delta m = 0.028697 \text{ u}$, or 0.7%. Therefore, the total amount of energy released in forming the helium nucleus is $E_b = \Delta mc^2 = 26.731 \text{ MeV}$. This is known as the **binding energy** of the helium nucleus. If the nucleus were to be broken apart into its constituent protons and neutrons, the amount of energy required to accomplish the task would be 26.731 MeV.

Example 10.3.2. Is this source of nuclear energy sufficient to power the Sun during its lifetime? For simplicity, assume also that the Sun was originally 100% hydrogen and that only the inner 10% of the Sun's mass becomes hot enough to convert hydrogen into helium.

Since 0.7% of the mass of hydrogen would be converted to energy in forming a helium nucleus, the amount of nuclear energy available in the Sun would be

$$E_{\text{nuclear}} = 0.1 \times 0.007 \times M_{\odot} c^2 = 1.3 \times 10^{44} \text{ J}.$$

continued

⁴The quantity A_j defined on page 292 is approximately equal to the mass number.

This gives a **nuclear timescale** of approximately

$$t_{\text{nuclear}} = \frac{E_{\text{nuclear}}}{L_{\odot}} \quad (10.25)$$

$$\sim 10^{10} \text{ yr},$$

more than enough time to account for the age of Moon rocks.

Quantum Mechanical Tunneling

Apparently, sufficient energy is available in the nuclei of atoms to provide a source for stellar luminosities, but can nuclear reactions actually occur in the interiors of stars? For a reaction to occur, the nuclei of atoms must collide, forming new nuclei in the process. However, all nuclei are positively charged, meaning that a Coulomb potential energy barrier must be overcome before contact can occur. Figure 10.4 shows the characteristic shape of the potential energy curve that an atomic nucleus would experience when approaching another nucleus. The curve is composed of two parts: The portion outside the nucleus is the potential energy that exists between two positively charged nuclei, and the portion inside the nucleus forms a *potential well* governed by the **strong nuclear force** that binds the nucleus together. The strong nuclear force is a very short-range force that acts between all nucleons within the atom. It is an attractive force that dominates the Coulomb repulsion between protons. Clearly, if such a force did not exist, a nucleus would immediately fly apart.

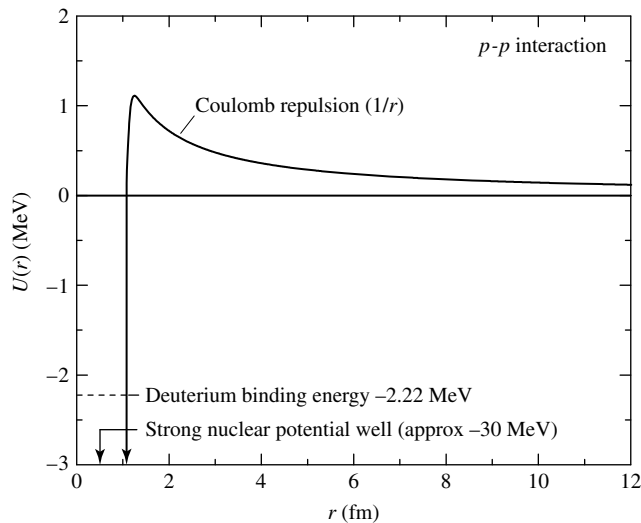


FIGURE 10.4 The potential energy curve characteristic of nuclear reactions. The Coulomb repulsion between positive nuclei results in a barrier that is inversely proportional to the separation between nuclei and is proportional to the product of their charges. The nuclear potential well inside the nucleus is due to the attractive strong nuclear force.

If we assume that the energy required to overcome the Coulomb barrier is provided by the thermal energy of the gas, and that all nuclei are moving nonrelativistically, then the temperature $T_{\text{classical}}$ required to overcome the barrier can be estimated. Since all of the particles in the gas are in random motion, it is appropriate to refer to the relative velocity v between two nuclei and their *reduced mass*, μ_m , given by Eq. (2.22) (note that we are not referring here to the mean molecular weight, μ). Equating the initial kinetic energy of the reduced mass to the potential energy of the barrier gives the position of the classical “turn-around point.” Now, using Eq. (10.17) yields

$$\frac{1}{2}\mu_m \overline{v^2} = \frac{3}{2}kT_{\text{classical}} = \frac{1}{4\pi\epsilon_0} \frac{Z_1 Z_2 e^2}{r},$$

where $T_{\text{classical}}$ denotes the temperature required for an average particle to overcome the barrier, Z_1 and Z_2 are the numbers of protons in each nucleus, and r is their distance of separation. Assuming that the radius of a typical nucleus is on the order of $1 \text{ fm} = 10^{-15} \text{ m}$, the temperature needed to overcome the Coulomb potential energy barrier is approximately

$$\begin{aligned} T_{\text{classical}} &= \frac{Z_1 Z_2 e^2}{6\pi\epsilon_0 k r} \\ &\sim 10^{10} \text{ K} \end{aligned} \tag{10.26}$$

for a collision between two protons ($Z_1 = Z_2 = 1$). However, the central temperature of the Sun is only $1.57 \times 10^7 \text{ K}$, much lower than required here. Even taking into consideration the fact that the Maxwell–Boltzmann distribution indicates that a significant number of particles have speeds well in excess of the average speed of particles in the gas, classical physics is unable to explain how a sufficient number of particles can overcome the Coulomb barrier to produce the Sun’s observed luminosity.

As was mentioned in Section 5.4, quantum mechanics tells us that it is never possible to know both the position and the momentum of a particle to unlimited accuracy. The Heisenberg uncertainty principle states that the uncertainties in position and momentum are related by

$$\Delta x \Delta p_x \geq \frac{\hbar}{2}.$$

The uncertainty in the position of one proton colliding with another may be so large that even though the kinetic energy of the collision is insufficient to overcome the classical Coulomb barrier, one proton might nevertheless find itself within the central potential well defined by the strong force of the other. This quantum mechanical tunneling has no classical counterpart (recall the discussion in Section 5.4). Of course, the greater the ratio of the potential energy barrier height to the particle’s kinetic energy or the wider the barrier, the less likely tunneling becomes.

As a crude estimate of the effect of tunneling on the temperature necessary to sustain nuclear reactions, assume that a proton must be within approximately one de Broglie wavelength of its target in order to tunnel through the Coulomb barrier. Recalling that the

wavelength of a massive particle is given by $\lambda = h/p$ (Eq. 5.17), rewriting the kinetic energy in terms of momentum,

$$\frac{1}{2}\mu_m v^2 = \frac{p^2}{2\mu_m},$$

and setting the distance of closest approach equal to one wavelength (where the potential energy barrier height is equal to the original kinetic energy) give

$$\frac{1}{4\pi\epsilon_0} \frac{Z_1 Z_2 e^2}{\lambda} = \frac{p^2}{2\mu_m} = \frac{(h/\lambda)^2}{2\mu_m}.$$

Solving for λ and substituting $r = \lambda$ into Eq. (10.26), we find the quantum mechanical estimate of the temperature required for a reaction to occur:

$$T_{\text{quantum}} = \frac{Z_1^2 Z_2^2 e^4 \mu_m}{12\pi^2 \epsilon_0^2 h^2 k}. \quad (10.27)$$

Again assuming the collision of two protons, $\mu_m = m_p/2$ and $Z_1 = Z_2 = 1$. Substituting, we find that $T_{\text{quantum}} \sim 10^7$ K. In this case, if we assume the effects of quantum mechanics, the temperature required for nuclear reactions is consistent with the estimated central temperature of the Sun.

Nuclear Reaction Rates and the Gamow Peak

Now that the possibility of a nuclear energy source has been established, we need a more detailed description of nuclear *reaction rates* in order to apply them to the development of stellar models. For instance, not all particles in a gas of temperature T will have sufficient kinetic energy and the necessary wavelength to tunnel through the Coulomb barrier successfully. Consequently, the reaction rate per energy interval must be described in terms of the number density of particles having energies within a specific range, combined with the probability that those particles can actually tunnel through the Coulomb barrier of the target nucleus. The total nuclear reaction rate is then integrated over all possible energies.

First consider the number density of nuclei within a specified energy interval. As we have seen, the Maxwell–Boltzmann distribution (Eq. 8.1) relates the number density of particles with velocities between v and $v + dv$ to the temperature of the gas. Assuming that particles are initially sufficiently far apart that the potential energy may be neglected, the nonrelativistic⁵ kinetic energy relation describes the total energy of the particles, or $K = E = \mu_m v^2/2$. Solving for the velocity and substituting, we can write the Maxwell–Boltzmann distribution in terms of the number of particles with kinetic energies between E and $E + dE$ as

$$n_E dE = \frac{2n}{\pi^{1/2}} \frac{1}{(kT)^{3/2}} E^{1/2} e^{-E/kT} dE \quad (10.28)$$

(the proof of this relation is left to Problem 10.6).

⁵In astrophysical processes, nuclei are usually nonrelativistic, except in the extreme environment of neutron stars. Because of the much smaller masses of electrons, it cannot be assumed that they are also nonrelativistic, however.

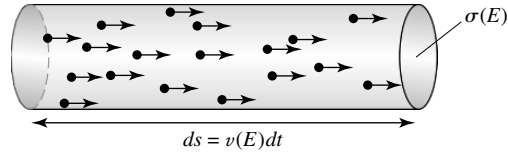


FIGURE 10.5 The number of reactions per unit time between particles of type i and a target x of cross section $\sigma(E)$ may be thought of in terms of the number of particles in a cylinder of cross-sectional area $\sigma(E)$ and length $ds = v(E)dt$ that will reach the target in a time interval dt .

Equation (10.28) gives the number of particles per unit volume that have energies in the range dE , but it does not describe the probability that particles will actually interact. To account for this factor, the idea of a **cross section** is re-introduced.⁶ Define the cross section $\sigma(E)$ to be the number of reactions per target nucleus per unit time, divided by the flux of incident particles, or

$$\sigma(E) \equiv \frac{\text{number of reactions/nucleus/time}}{\text{number of incident particles/area/time}}.$$

Although $\sigma(E)$ is strictly a measure of probability, as mentioned in Chapter 9, it can be thought of as *roughly* the cross-sectional area of the target particle; any incoming particle that strikes within that area, centered on the target, will result in a nuclear reaction.

To find the reaction rate in units of reactions volume⁻¹ time⁻¹, consider the number of particles that will hit a target of cross-sectional area $\sigma(E)$, assuming that all of the incident particles are moving in one direction. Let x denote a target particle and i denote an incident particle. If the number of incident particles per unit volume having energies between E and $E + dE$ is $n_{iE} dE$, then the number of reactions, dN_E , is the number of particles that can strike x in a time interval dt with a velocity $v(E) = \sqrt{2E/\mu_m}$.

The number of incident particles is just the number contained within a cylinder of volume $\sigma(E)v(E)dt$ (see Fig. 10.5), or

$$dN_E = \sigma(E)v(E)n_{iE}dE dt.$$

Now, the number of incident particles per unit volume with the appropriate velocity (or kinetic energy) is some fraction of the total number of particles in the sample,

$$n_{iE}dE = \frac{n_i}{n}n_EdE,$$

where $n_i = \int_0^\infty n_{iE}dE$, $n = \int_0^\infty n_EdE$, and n_EdE is given by Eq. (10.28). Therefore, the number of reactions per target nucleus per time interval dt having energies between E and $E + dE$ is

$$\frac{\text{reactions per nucleus}}{\text{time interval}} = \frac{dN_E}{dt} = \sigma(E)v(E)\frac{n_i}{n}n_EdE.$$

⁶The concept of a cross section was first discussed in the context of determining the mean free path between collisions in Section 9.2.

Finally, if there are n_x targets per unit volume, the total number of reactions per unit volume per unit time, integrated over all possible energies, is

$$r_{ix} = \int_0^\infty n_x n_i \sigma(E) v(E) \frac{n_E}{n} dE. \quad (10.29)$$

To evaluate Eq. (10.29) we must know the functional form of $\sigma(E)$. Unfortunately, $\sigma(E)$ changes rapidly with energy, and its functional form is complicated. It is also important to compare $\sigma(E)$ with experimental data. However, stellar thermal energies are quite low compared to energies found in laboratory experimentation, and significant extrapolation is usually required to obtain comparison data for stellar nuclear reaction rates.

The process of determining $\sigma(E)$ can be improved somewhat if the terms most strongly dependent on energy are factored out first. We have already suggested that the cross section can be roughly thought of as being a physical area. Moreover, the size of a nucleus, measured in terms of its ability to “touch” target nuclei, is approximately one de Broglie wavelength in radius ($r \sim \lambda$). Combining these ideas, the cross section of the nucleus $\sigma(E)$ should be proportional to

$$\sigma(E) \propto \pi \lambda^2 \propto \pi \left(\frac{h}{p} \right)^2 \propto \frac{1}{E}.$$

To obtain the last expression, we have again used the nonrelativistic relation, $K = E = \mu_m v^2/2 = p^2/2\mu_m$.

We have also mentioned previously that the ability to tunnel through the Coulomb barrier is related to the ratio of the barrier height to the initial kinetic energy of the incoming nucleus, a factor that must be considered in the cross section. If the barrier height U_c is zero, the probability of successfully penetrating it necessarily equals one (100%). As the barrier height increases relative to the initial kinetic energy of the incoming nucleus, the probability of penetration must decrease, asymptotically approaching zero as the potential energy barrier height goes to infinity. In fact, the tunneling probability is exponential in nature. Since $\sigma(E)$ must be related to the tunneling probability, we have

$$\sigma(E) \propto e^{-2\pi^2 U_c/E}. \quad (10.30)$$

The factor of $2\pi^2$ arises from the strict quantum mechanical treatment of the problem. Again assuming that $r \sim \lambda = h/p$, taking the ratio of the barrier potential height U_c to particle kinetic energy E gives

$$\frac{U_c}{E} = \frac{Z_1 Z_2 e^2 / 4\pi \epsilon_0 r}{\mu_m v^2 / 2} = \frac{Z_1 Z_2 e^2}{2\pi \epsilon_0 h v}.$$

After some manipulation, we find that

$$\sigma(E) \propto e^{-bE^{-1/2}}, \quad (10.31)$$

where

$$b \equiv \frac{\pi \mu_m^{1/2} Z_1 Z_2 e^2}{2^{1/2} \epsilon_0 h}.$$

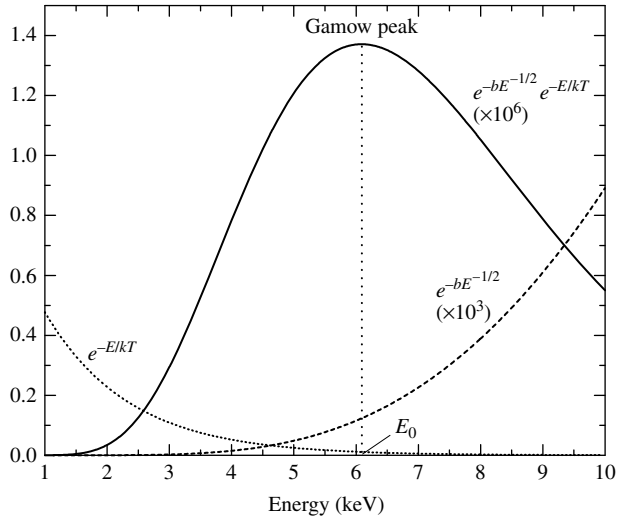


FIGURE 10.6 The likelihood that a nuclear reaction will occur is a function of the kinetic energy of the collision. The Gamow peak arises from the contribution of the $e^{-E/kT}$ Maxwell-Boltzmann high-energy tail and the $e^{-bE^{-1/2}}$ Coulomb barrier penetration term. This particular example represents the collision of two protons at the central temperature of the Sun. (Note that $e^{-bE^{-1/2}}$ and $e^{-E/kT}$ have been multiplied by 10^3 and 10^6 , respectively, to more readily illustrate the functional dependence on energy.)

Clearly, b depends on the masses and electric charges of the two nuclei involved in the interaction.

Combining the previous results and defining $S(E)$ to be some (we hope) slowly varying function of energy, we may now express the cross section as⁷

$$\sigma(E) = \frac{S(E)}{E} e^{-bE^{-1/2}}. \quad (10.32)$$

Substituting Eqs. (10.28) and (10.32) into Eq. (10.29) and simplifying, the reaction rate integral becomes

$$r_{ix} = \left(\frac{2}{kT} \right)^{3/2} \frac{n_i n_x}{(\mu_m \pi)^{1/2}} \int_0^\infty S(E) e^{-bE^{-1/2}} e^{-E/kT} dE. \quad (10.33)$$

In Eq. (10.33), the term $e^{-E/kT}$ represents the high-energy wing of the Maxwell-Boltzmann distribution, and the term $e^{-bE^{-1/2}}$ comes from the penetration probability. As can be seen in Fig. 10.6, the product of these two factors produces a strongly peaked curve, known as the **Gamow peak** after George Gamow (1904–1968), the physicist who first

⁷The angular momentum of the interacting particles also plays a role in nuclear reaction rates, but it is generally a minor component for reactions of astrophysical significance.

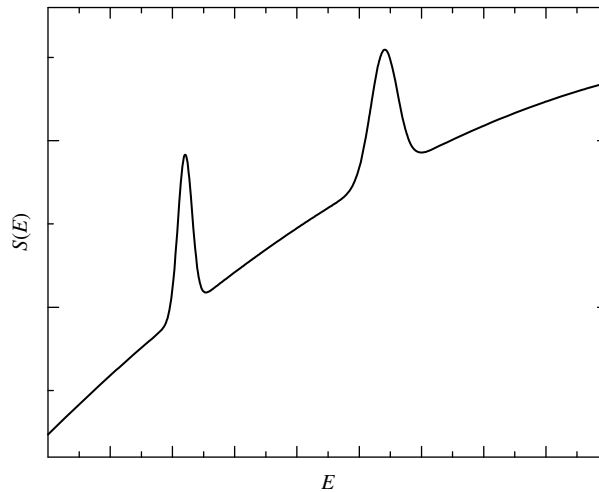


FIGURE 10.7 A hypothetical example of the effect of resonance on $S(E)$.

investigated Coulomb barrier penetration. The top of the curve occurs at the energy

$$E_0 = \left(\frac{bkT}{2} \right)^{2/3}. \quad (10.34)$$

As a consequence of the Gamow peak, the greatest contribution to the reaction rate integral comes in a fairly narrow energy band that depends on the temperature of the gas, together with the charges and masses of the constituents of the reaction.

Assuming that $S(E)$ is indeed slowly varying across the Gamow peak, it may be approximated by its value at E_0 [$S(E) \simeq S(E_0) = \text{constant}$] and removed from inside of the integral. Also, it is generally much easier to extrapolate laboratory results if they are expressed in terms of $S(E)$.

Resonance

In some cases, however, $S(E)$ can vary quite rapidly, peaking at specific energies, as illustrated schematically in Fig. 10.7. These energies correspond to energy levels within the nucleus, analogous to the orbital energy levels of electrons. It is a **resonance** between the energy of the incoming particle and differences in energy levels within the nucleus that accounts for these strong peaks. A detailed discussion of these resonance peaks is beyond the scope of this book.⁸

Electron Screening

Yet another factor influencing reaction rates is **electron screening**. On average, the electrons liberated when atoms are ionized at the high temperatures of stellar interiors produce a

⁸See Clayton (1983) or Arnett (1996) for excellent and detailed discussions of resonance peaks.

“sea” of negative charge that partially hides the target nucleus, reducing its *effective* positive charge. The result of this reduced positive charge is a lower Coulomb barrier to the incoming nucleus and an enhanced reaction rate. By including electron screening, the effective Coulomb potential becomes

$$U_{\text{eff}} = \frac{1}{4\pi\epsilon_0} \frac{Z_1 Z_2 e^2}{r} + U_s(r),$$

where $U_s(r) < 0$ is the electron screening contribution. Electron screening can be significant, sometimes enhancing the helium-producing reactions by 10% to 50%.

Representing Nuclear Reaction Rates Using Power Laws

It is often illuminating to write the complicated reaction rate equations in the form of a power law centered at a particular temperature. Neglecting the screening factor, in the case of a two-particle interaction, the reaction rate would become

$$r_{ix} \simeq r_0 X_i X_x \rho^{\alpha'} T^{\beta},$$

where r_0 is a constant, X_i and X_x are the mass fractions of the two particles, and α' and β are determined from the power law expansion of the reaction rate equations. Usually $\alpha' = 2$ for a two-body collision, and β can range from near unity to 40 or more.

By combining the reaction rate equation with the amount of energy released per reaction, we can calculate the amount of energy released per second in each kilogram of stellar material. If $\mathcal{E}_0$ is the amount of energy released per reaction, the amount of energy liberated per kilogram of material per second becomes

$$\epsilon_{ix} = \left(\frac{\mathcal{E}_0}{\rho} \right) r_{ix},$$

or, in the form of a power law,

$$\epsilon_{ix} = \epsilon'_0 X_i X_x \rho^{\alpha} T^{\beta}, \quad (10.35)$$

where $\alpha = \alpha' - 1$. ϵ_{ix} has units of W kg^{-1} and the sum of ϵ_{ix} for all reactions is the total nuclear energy generation rate. This form of the nuclear energy generation rate will be used later to show the dependence of energy production on temperature and density for several reaction sequences typically operating in stellar interiors.

The Luminosity Gradient Equation

To determine the luminosity of a star, we must now consider all of the energy generated by stellar material. The contribution to the total luminosity due to an infinitesimal mass dm is simply

$$dL = \epsilon dm,$$

where ϵ is the *total* energy released per kilogram per second by all nuclear reactions and by gravity, or $\epsilon = \epsilon_{\text{nuclear}} + \epsilon_{\text{gravity}}$. It is worth noting that $\epsilon_{\text{gravity}}$ could be negative if the

star is expanding, a point to be discussed later. For a spherically symmetric star, the mass of a thin shell of thickness dr is just $dm = dM_r = \rho dV = 4\pi r^2 \rho dr$ (recall Fig. 10.2). Substituting and dividing by the shell thickness, we have

$$\boxed{\frac{dL_r}{dr} = 4\pi r^2 \rho \epsilon,} \quad (10.36)$$

where L_r is the *interior luminosity* due to all of the energy generated within the star's interior out to the radius r . Equation (10.36) is another of the fundamental stellar structure equations.

Stellar Nucleosynthesis and Conservation Laws

The remaining problem in understanding nuclear reactions is the exact sequence of steps by which one element is converted into another, a process known as **nucleosynthesis**. Our estimate of the nuclear timescale for the Sun was based on the assumption that four hydrogen nuclei are converted into helium. However, it is highly unlikely that this could occur via a four-body collision (i.e., all nuclei hitting simultaneously). For the process to occur, the final product must be created by a chain of reactions, each involving much more probable two-body interactions. In fact, we derived the reaction rate equation under the assumption that only two nuclei would collide at any one time.

The process by which a chain of nuclear reactions leads to the final product cannot happen in a completely arbitrary way, however; a series of particle *conservation laws* must be obeyed. In particular, during every reaction it is necessary to conserve electric charge, the number of nucleons, and the number of leptons. The term **lepton** means a “light thing” and includes electrons, positrons, neutrinos, and antineutrinos.

Although antimatter is extremely rare in comparison with matter, it plays an important role in subatomic physics, including nuclear reactions. Antimatter particles are identical to their matter counterparts but have opposite attributes, such as electric charge. Antimatter also has the characteristic (often used in science fiction) that a collision with its matter counterpart results in complete annihilation of both particles, accompanied by the production of energetic photons. For instance,

$$e^- + e^+ \rightarrow 2\gamma,$$

where e^- , e^+ , and γ denote an electron, positron, and photon, respectively. Note that two photons are required to conserve both momentum and energy simultaneously.

Neutrinos and antineutrinos (symbolized by ν and $\bar{\nu}$, respectively) are an interesting class of particles in their own right and will be discussed often in the remainder of this text.⁹ Neutrinos are electrically neutral and have a very small but non-zero mass ($m_\nu < 2.2 \text{ eV}/c^2$). One of the interesting characteristics of a neutrino is its extremely small cross section for interactions with other matter, making it very difficult to detect. Typically

⁹These particles were originally proposed by Wolfgang Pauli in 1930, in order that energy and momentum might be conserved in certain reaction processes. In 1934, they were given the name *neutrinos* (“little neutral ones”) by Italian physicist Enrico Fermi (1901–1954).

$\sigma_\nu \sim 10^{-48} \text{ m}^2$, implying that at densities common to stellar interiors, a neutrino's mean free path is on the order of $10^{18} \text{ m} \sim 10 \text{ pc}$, or nearly $10^9 R_\odot$! After being produced in the deep interior, neutrinos almost always successfully escape from the star. One exception to this transparency of stellar material to neutrinos occurs with important consequences during a supernova explosion, as will be discussed in Chapter 15.

Since electrons and positrons have charges equal in magnitude to that of a proton, these leptons will contribute to the charge conservation requirement while their total **lepton numbers** must also be conserved. Note that in counting the number of leptons involved in a nuclear reaction, we treat matter and antimatter differently. Specifically, the total number of matter leptons *minus* the total number of antimatter leptons must remain constant.

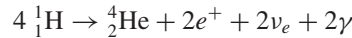
To assist in counting the number of nucleons and the total electric charge, nuclei will be represented in this text by the symbol

$${}_Z^AX,$$

where X is the chemical symbol of the element (H for hydrogen, He for helium, etc.), Z is the number of protons (the total positive charge, in units of e), and A is the mass number (the total number of nucleons, protons plus neutrons).¹⁰

The Proton–Proton Chains

Applying the conservation laws, one chain of reactions that can convert hydrogen into helium is the first **proton–proton chain** (PPI). It involves a reaction sequence that ultimately results in



through the intermediate production of deuterium (${}_1^2\text{H}$) and helium-3 (${}_2^3\text{He}$). The entire **PPI** reaction chain is¹¹



Each step of the PPI chain has its own reaction rate, since different Coulomb barriers and cross sections are involved. The slowest step in the sequence is the initial one, because it involves the *decay* of a proton into a neutron via $p^+ \rightarrow n + e^+ + \nu_e$. Such a decay involves the **weak force**, another of the four known forces.¹²

¹⁰Since an element is uniquely determined by the number of protons (Z) in the nucleus, specifying both X and Z is redundant. As a result, some texts use the less cumbersome notation AX . However, this notation makes it more difficult to keep track of the electric charge in a nuclear reaction.

¹¹Approximately 0.4% of the time, the first reaction step is accomplished by the so-called pep reaction: ${}_1^1\text{H} + e^- + {}_1^1\text{H} \rightarrow {}_1^2\text{H} + \nu_e$.

¹²Each of the four forces has now been mentioned: the gravitational force, which involves all particles with mass–energy; the electromagnetic force, associated with photons and electric charge; the strong force that binds nuclei together; and the weak force of radioactive beta (electron/positron) decay.

The production of helium-3 nuclei in the PP I chain also provides for the possibility of their interaction directly with helium-4 nuclei, resulting in a second branch of the proton–proton chain. In an environment characteristic of the center of the Sun, 69% of the time a helium-3 interacts with another helium-3 in the PP I chain, whereas 31% of the time the **PP II** chain occurs:



Yet another branch, the **PP III** chain, is possible because the capture of an electron by the beryllium-7 nucleus in the PP II chain competes with the capture of a proton (a proton is captured only 0.3% of the time in the center of the Sun):



The three branches of the proton–proton (pp) chain, along with their branching ratios, are summarized in Fig. 10.8.

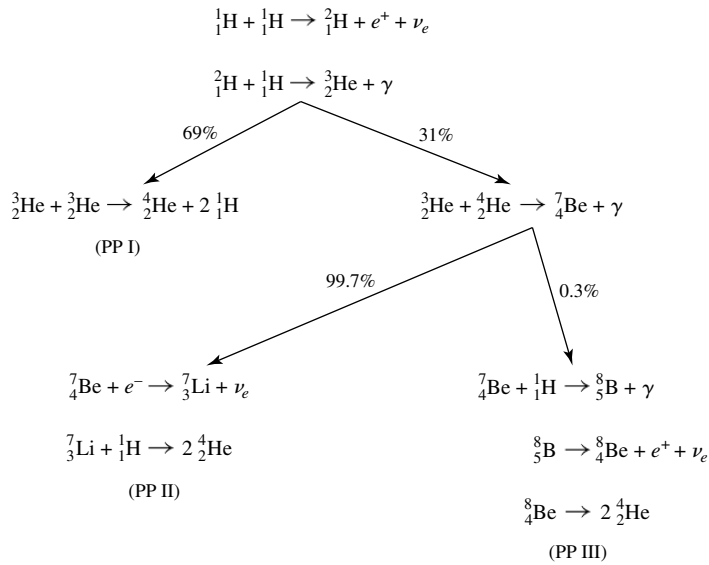


FIGURE 10.8 The three branches of the pp chain, along with the branching ratios appropriate for conditions in the core of the Sun.

Beginning with Eq. (10.33), the nuclear energy generation rate for the combined pp chain is calculated to be

$$\epsilon_{pp} = 0.241 \rho X^2 f_{pp} \psi_{pp} C_{pp} T_6^{-2/3} e^{-33.80 T_6^{-1/3}} \text{ W kg}^{-1}, \quad (10.46)$$

where T_6 is a dimensionless expression of temperature in units of 10^6 K (or $T_6 \equiv T/10^6$ K). $f_{pp} = f_{pp}(X, Y, \rho, T) \simeq 1$ is the pp chain screening factor, $\psi_{pp} = \psi_{pp}(X, Y, T) \simeq 1$ is a correction factor that accounts for the simultaneous occurrence of PP I, PP II, and PP III, and $C_{pp} \simeq 1$ involves higher-order correction terms.¹³

When written as a power law (e.g., Eq. 10.35) near $T = 1.5 \times 10^7$ K, the energy generation rate has the form

$$\epsilon_{pp} \simeq \epsilon'_{0,pp} \rho X^2 f_{pp} \psi_{pp} C_{pp} T_6^4, \quad (10.47)$$

where $\epsilon'_{0,pp} = 1.08 \times 10^{-12} \text{ W m}^3 \text{ kg}^{-2}$. The power law form of the energy generation rate demonstrates a relatively modest temperature dependence of T^4 near $T_6 = 15$.

The CNO Cycle

A second, independent cycle also exists for the production of helium-4 from hydrogen. This cycle was proposed by Hans Bethe (1906–2005) in 1938, just six years after the discovery of the neutron. In the **CNO cycle**, carbon, nitrogen, and oxygen are used as catalysts, being consumed and then regenerated during the process. Just as with the pp chain, the CNO cycle has competing branches. The first branch culminates with the production of carbon-12 and helium-4:



The second branch occurs only about 0.04% of the time and arises when the last reaction (Eq. 10.53) produces oxygen-16 and a photon, rather than carbon-12 and helium-4:



¹³Expressions for the various correction terms are given in the stellar structure code `StatStar`, described in Appendix L.

The energy generation rate for the CNO cycle is given by

$$\epsilon_{\text{CNO}} = 8.67 \times 10^{20} \rho X X_{\text{CNO}} C_{\text{CNO}} T_6^{-2/3} e^{-152.28 T_6^{-1/3}} \text{ W kg}^{-1}, \quad (10.58)$$

where X_{CNO} is the total mass fraction of carbon, nitrogen, and oxygen, and C_{CNO} is a higher-order correction term. When written as a power law centered about $T = 1.5 \times 10^7$ K (see Eq. 10.35), this energy equation becomes

$$\epsilon_{\text{CNO}} \simeq \epsilon'_{0,\text{CNO}} \rho X X_{\text{CNO}} T_6^{19.9}, \quad (10.59)$$

where $\epsilon'_{0,\text{CNO}} = 8.24 \times 10^{-31} \text{ W m}^3 \text{ kg}^{-2}$. As shown by the power law dependence, the CNO cycle is much more strongly temperature-dependent than is the pp chain. This property implies that low-mass stars, which have smaller central temperatures, are dominated by the pp chains during their “hydrogen burning” evolution, whereas more massive stars, with their higher central temperatures, convert hydrogen to helium by the CNO cycle. The transition in stellar mass between stars dominated by the pp chain and those dominated by the CNO cycle occurs for stars slightly more massive than our Sun. This difference in nuclear reaction processes plays an important role in the structure of stellar interiors, as will be seen in the next section.

When hydrogen is converted into helium by either the pp chain or the CNO cycle, the mean molecular weight μ of the gas increases. If neither the temperature nor the density of the gas changes, the ideal gas law predicts that the central pressure will necessarily decrease. As a result, the star would no longer be in hydrostatic equilibrium and would begin to collapse. This collapse has the effect of actually raising both the temperature and the density to compensate for the increase in μ (recall the virial theorem, Eq. 2.46). When the temperature and density become sufficiently high, helium nuclei can overcome their Coulomb repulsion and begin to “burn.”

The Triple Alpha Process of Helium Burning

The reaction sequence by which helium is converted into carbon is known as the **triple alpha process**. The process takes its name from the historical result that the mysterious alpha particles detected in some types of radioactive decay were shown by Rutherford to be helium-4 (${}^4\text{He}$) nuclei. The triple alpha process is



In the triple alpha process, the first step produces an unstable beryllium nucleus that will rapidly decay back into two separate helium nuclei if not immediately struck by another alpha particle. As a result, this reaction may be thought of as a three-body interaction, and therefore, the reaction rate depends on $(\rho Y)^3$. The nuclear energy generation rate is given by

$$\epsilon_{3\alpha} = 50.9 \rho^2 Y^3 T_8^{-3} f_{3\alpha} e^{-44.027 T_8^{-1}} \text{ W kg}^{-1}, \quad (10.62)$$

where $T_8 \equiv T/10^8$ K and $f_{3\alpha}$ is the screening factor for the triple alpha process. Written as a power law centered on $T = 10^8$ K (see Eq. 10.35), it demonstrates a very dramatic temperature dependence:

$$\epsilon_{3\alpha} \simeq \epsilon'_{0,3\alpha} \rho^2 Y^3 f_{3\alpha} T_8^{41.0}. \quad (10.63)$$

With such a strong dependence, even a small increase in temperature will produce a large increase in the amount of energy generated per second. For instance, an increase of only 10% in temperature raises the energy output rate by more than 50 times!

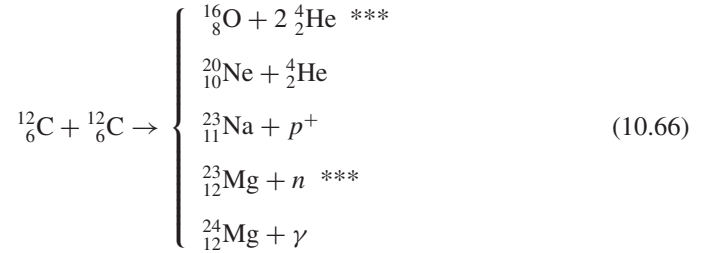
Carbon and Oxygen Burning

In the high-temperature environment of helium burning, other competing processes are also at work. After sufficient carbon has been generated by the triple alpha process, it becomes possible for carbon nuclei to capture alpha particles, producing oxygen. Some of the oxygen in turn can capture alpha particles to produce neon.

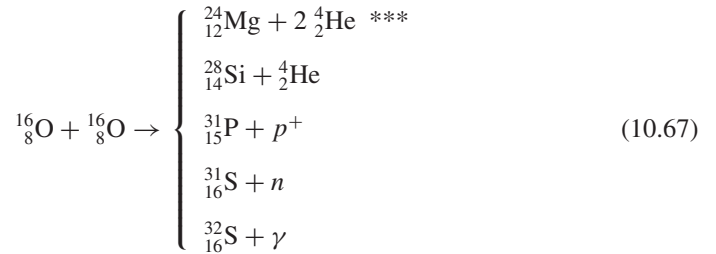


At helium-burning temperatures, the continued capture of alpha particles leading to progressively more massive nuclei quickly becomes prohibitive due to the ever higher Coulomb barrier.

If a star is sufficiently massive, still higher central temperatures can be obtained and many other nuclear products become possible. Examples of available reactions include carbon burning reactions near 6×10^8 K,



and oxygen burning reactions near 10^9 K,



Reactions marked by *** are ones for which energy is absorbed rather than released and are referred to as being **endothermic**; energy-releasing reactions are **exothermic**. In endothermic reactions the product nucleus actually possesses more energy per nucleon than did the nuclei from which it formed. Such reactions occur at the expense of the energy released by exothermic reactions or by gravitational collapse (the virial theorem). In general, endothermic reactions are much less likely to occur than exothermic reactions under conditions that normally prevail in stellar interiors.

The Binding Energy Per Nucleon

A useful quantity in understanding the energy release in nuclear reactions is the binding energy per nucleon, E_b/A , where

$$E_b = \Delta mc^2 = [Zm_p + (A - Z)m_n - m_{\text{nucleus}}] c^2.$$

Figure 10.9 shows E_b/A versus the mass number. It is apparent that for relatively small values of A (less than 56), several nuclei have abnormally high values of E_b/A relative to others of similar mass. Among these unusually stable nuclei are ${}^4_2\text{He}$ and ${}^{16}_8\text{O}$, which, along with ${}^1_1\text{H}$, are the most abundant nuclei in the universe. This unusual stability arises from an inherent shell structure of the nucleus, analogous to the shell structure of atomic energy levels that accounts for the chemical nature of elements. These unusually stable nuclei are called *magic nuclei*.

It is believed that shortly after the Big Bang the early universe was composed primarily of hydrogen and helium, with no heavy elements. Today, Earth and its inhabitants contain an abundance of heavier metals. The study of stellar nucleosynthesis strongly suggests that

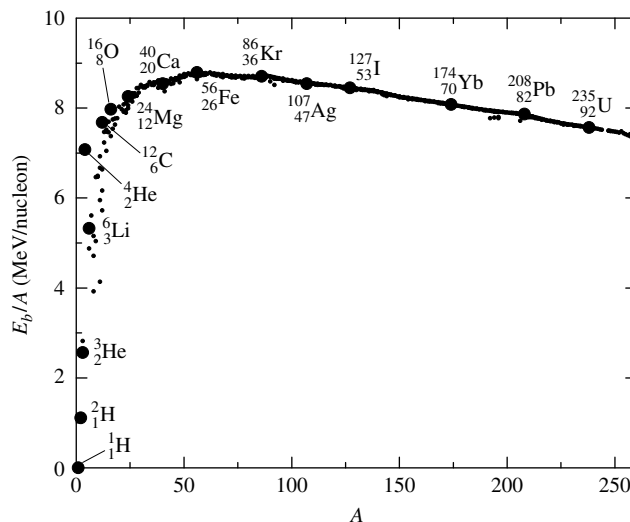


FIGURE 10.9 The binding energy per nucleon, E_b/A , as a function of mass number, A . Notice that several nuclei, most notably ${}^4_2\text{He}$ (see also ${}^{12}_6\text{C}$ and ${}^{16}_8\text{O}$), lie well above the general trend of the other nuclei, indicating unusual stability. At the peak of the curve is ${}^{56}_{26}\text{Fe}$, the most stable of all nuclei.

these heavier nuclei were generated in the interiors of stars. It can be said that we are all “star dust,” the product of heavy element generation within previous generations of stars.

Another important feature of Fig. 10.9 is the broad peak around $A = 56$. At the top of the peak is an isotope of iron, ${}^{56}_{26}\text{Fe}$, the most stable of all nuclei. As successively more massive nuclei are created in stellar interiors, the iron peak is approached from the left. These fusion reactions result in the liberation of energy.¹⁴ Consequently, the ultimate result of successive chains of nuclear reactions within stars is the production of iron, assuming sufficient energy is available to overcome the Coulomb barrier. If a star is massive enough to create the central temperatures and densities necessary to produce iron, the results are spectacular, as we will see in Chapter 15.

Considering what we have learned in this section about stellar nucleosynthesis, it should come as no surprise that the most abundant nuclear species in the cosmos are, in order, ${}^1_1\text{H}$, ${}^4_2\text{He}$, ${}^{16}_8\text{O}$, ${}^{12}_6\text{C}$, ${}^{20}_{10}\text{Ne}$, ${}^{14}_7\text{N}$, ${}^{24}_{12}\text{Mg}$, ${}^{28}_{14}\text{Si}$, and ${}^{56}_{26}\text{Fe}$ (recall Table 9.2).¹⁵ The abundances are the result of the dominant nuclear reaction processes that occur in stars, together with the nuclear configurations that result in the most stable nuclei. Reactions capable of producing the diversity of other isotopes are discussed in Section 15.3.

10.4 ■ ENERGY TRANSPORT AND THERMODYNAMICS

One stellar structure equation remains to be developed. We have already related the fundamental quantities P , M , and L to the independent variable r through differential equations that describe hydrostatic equilibrium, mass conservation, and energy generation, respectively; see Eqs. (10.6), (10.7), and (10.36). However, we have not yet found a differential equation relating the basic parameter of temperature, T , to r . Moreover, we have not explicitly developed equations that describe the processes by which heat generated via nuclear reactions or gravitational contraction is carried from the deep interior to the surface of the star.

Three Energy Transport Mechanisms

Three different energy transport mechanisms operate in stellar interiors. **Radiation** allows the energy produced by nuclear reactions and gravitation to be carried to the surface via photons, the photons being absorbed and re-emitted in nearly random directions as they encounter matter (recall the discussion in Section 9.3). This suggests that the opacity of the material must play an important role, as one would expect. **Convection** can be a very efficient transport mechanism in many regions of a star, with hot, buoyant mass elements carrying excess energy outward while cool elements fall inward. Finally, **conduction** transports heat via collisions between particles. Although conduction can play an important role in some stellar environments, it is generally insignificant in most stars throughout the majority of their lifetimes and will not be discussed further here.

¹⁴Energy is also released when the peak is approached from the right via fission reactions that produce nuclei of smaller mass, again resulting in more stable nuclei. This type of reaction process is important in the fission reactors of nuclear power plants.

¹⁵The relative abundances of the elements will be discussed further in Section 15.3; see Fig. 15.16.

The Radiative Temperature Gradient

First consider radiation transport. In Chapter 9 we found that the radiation pressure gradient is given by Eq. (9.31),

$$\frac{dP_{\text{rad}}}{dr} = -\frac{\bar{\kappa}\rho}{c}F_{\text{rad}},$$

where F_{rad} is the outward radiative flux. However, from Eq. (10.19), the radiation pressure gradient may also be expressed as

$$\frac{dP_{\text{rad}}}{dr} = \frac{4}{3}aT^3\frac{dT}{dr}.$$

Equating the two expressions, we have

$$\frac{dT}{dr} = -\frac{3}{4ac}\frac{\bar{\kappa}\rho}{T^3}F_{\text{rad}}.$$

Finally, if we use the expression for the radiative flux (Eq. 3.2), written in terms of the local radiative luminosity of the star at radius r ,

$$F_{\text{rad}} = \frac{L_r}{4\pi r^2},$$

the temperature gradient for radiative transport becomes

$$\boxed{\frac{dT}{dr} = -\frac{3}{4ac}\frac{\bar{\kappa}\rho}{T^3}\frac{L_r}{4\pi r^2}}. \quad (10.68)$$

As either the flux or the opacity increases, the temperature gradient must become steeper (more negative) if radiation is to transport all of the required luminosity outward. The same situation holds as the density increases or the temperature decreases.

The Pressure Scale Height

If the temperature gradient becomes too steep, convection can begin to play an important role in the transport of energy. Physically, convection involves mass motions: hot parcels of matter move upward as cooler, denser parcels sink. Unfortunately, convection is a much more complex phenomenon than radiation at the macroscopic level. In fact, no truly satisfactory prescription yet exists to describe it adequately in stellar environments. *Fluid mechanics*, the field of physics describing the motion of gases and liquids, relies on a complicated set of three-dimensional equations known as the *Navier–Stokes equations*. However, at present, due in large part to the current limitations in computing power,¹⁶ most stellar structure codes are one-dimensional (i.e., depend on r only). It becomes necessary, therefore, to approximate an explicitly three-dimensional process by a one-dimensional phenomenological

¹⁶This limitation is being overcome to some extent with the development of ever faster computers with more memory, and through the implementation of more sophisticated numerical techniques.

theory. To complicate the situation even more, when convection is present in a star, it is generally quite turbulent, requiring a detailed understanding of the amount of viscosity (fluid friction) and heat dissipation involved. Also, a characteristic length scale for convection, typically referred to in terms of the **pressure scale height**, is often comparable to the size of the star. Lastly, the timescale for convection, taken to be the amount of time required for a convective element to travel a characteristic distance, is in some cases approximately equal to the timescale for changes in the structure of the star, implying that convection is strongly coupled to the star's dynamic behavior. The impact of these complications on the behavior of the star is not yet fundamentally understood.

The situation is not completely hopeless, however. Despite the difficulties encountered in attempting to treat stellar convection exactly, approximate (and even reasonable) results can usually be obtained. To estimate the size of a convective region in a star, consider the pressure scale height, H_P , defined as

$$\frac{1}{H_P} \equiv -\frac{1}{P} \frac{dP}{dr}. \quad (10.69)$$

If we assume for the moment that H_P is a constant, we can solve for the variation of pressure with radius, giving

$$P = P_0 e^{-r/H_P}.$$

Obviously, if $r = H_P$, then $P = P_0 e^{-1}$, so that H_P is the distance over which the gas pressure decreases by a factor of e . To find a convenient general expression for H_P , recall that from the equation for hydrostatic equilibrium (Eq. 10.6), $dP/dr = -\rho g$, where $g = GM_r/r^2$ is the local acceleration of gravity. Substituting into Eq. (10.69), the pressure scale height is simply

$$H_P = \frac{P}{\rho g}. \quad (10.70)$$

Example 10.4.1. To estimate a typical value for the pressure scale height in the Sun, assume that $\bar{P} = P_c/2$, where P_c is the central pressure, $\bar{\rho}_\odot$ is the average solar density, and

$$\bar{g} = \frac{G(M_\odot/2)}{(R_\odot/2)^2} = 550 \text{ m s}^{-2}.$$

Then we have

$$H_P \simeq 1.8 \times 10^8 \text{ m} \sim R_\odot/4.$$

Detailed calculations show that $H_P \sim R_\odot/10$ is more typical.

Internal Energy and the First Law of Thermodynamics

Understanding convective heat transport in stars, even in an approximate way, begins with some knowledge of thermodynamics. In the study of heat transport, conservation of energy

is expressed by **the first law of thermodynamics**,

$$\boxed{dU = dQ - dW}, \quad (10.71)$$

where the change in the internal energy of a mass element dU is given by the amount of heat *added* to that element, dQ , minus the work done *by* that element on its surroundings, dW . Throughout our discussion we will assume that these energy changes are measured *per unit mass*.

The internal energy of a system U is a **state function**, meaning that its value depends only on the present conditions of the gas, not on the history of any changes leading to its current state. Consequently, dU is independent of the actual process involved in the change. On the other hand, neither heat nor work is a state function. The amount of heat added to a system or the amount of work done by a system depends on the ways in which the processes are carried out. dQ and dW are referred to as *inexact differentials*, reflecting their path dependence.

Consider an ideal monatomic gas, a gas composed of single particles with no ionization. The total internal energy per unit mass is given by

$$\begin{aligned} U &= (\text{average energy/particle}) \times (\text{number of particles/mass}) \\ &= \bar{K} \times \frac{1}{\bar{m}} \end{aligned}$$

where $\bar{m} = \mu m_H$ is the average mass of a single particle in the gas. For an ideal gas, $\bar{K} = 3kT/2$ and the internal energy is given by

$$U = \frac{3}{2} \left(\frac{k}{\mu m_H} \right) T = \frac{3}{2} n R T, \quad (10.72)$$

where n is the number of moles¹⁷ *per unit mass*, $R = 8.314472 \text{ J mole}^{-1} \text{ K}^{-1}$ is the universal gas constant,¹⁸ and

$$nR = \frac{k}{\mu m_H}.$$

Clearly $U = U(\mu, T)$ is a function of the composition of the gas and its temperature. In this case of an ideal monatomic gas, the internal energy is just the kinetic energy per unit mass.

Specific Heats

The change in heat of the mass element dQ is generally expressed in terms of the **specific heat** C of the gas. The specific heat is defined as the amount of heat required to raise the

¹⁷ 1 mole = N_A particles, where $N_A = 6.02214199 \times 10^{23}$ is Avogadro's number, defined as the number of $^{12}_6\text{C}$ atoms required to produce exactly 12 grams of a pure sample.

¹⁸ $R = N_A k$.

temperature of a unit mass of a material by a unit temperature interval, or

$$C_P \equiv \left. \frac{\partial Q}{\partial T} \right|_P \quad \text{and} \quad C_V \equiv \left. \frac{\partial Q}{\partial T} \right|_V,$$

where C_P and C_V are the specific heats at constant pressure and volume, respectively.

Consider next the amount of work per unit mass, dW , done by the gas on its surroundings. Suppose that a cylinder of cross-sectional area A is filled with a gas of mass m and pressure P . The gas then exerts a force $F = PA$ on an end of the cylinder. If the end of the cylinder is a piston that moves through a distance dr , the work *per unit mass* performed by the gas may be expressed as

$$dW = \left(\frac{F}{m} \right) dr = \left(\frac{PA}{m} \right) dr = P dV,$$

V being defined as the **specific volume**, the volume *per unit mass*, or $V \equiv 1/\rho$. The first law of thermodynamics may now be expressed in the useful form

$$dU = dQ - P dV. \quad (10.73)$$

At constant volume, $dV = 0$, which gives $dU = dU|_V = dQ|_V$, or

$$dU = \left. \frac{\partial Q}{\partial T} \right|_V dT = C_V dT. \quad (10.74)$$

[It is important to note that because dU is independent of any specific process, the second equality of Eq. (10.74) is always valid, regardless of the type of thermodynamic process involved.] But from Eq. (10.72), $dU = (3nR/2) dT$ for a monatomic gas. Thus

$$C_V = \frac{3}{2}nR. \quad (10.75)$$

To find C_P for a monatomic gas, note that

$$dU = \left. \frac{\partial Q}{\partial T} \right|_P dT - P \left. \frac{\partial V}{\partial T} \right|_P dT. \quad (10.76)$$

In addition, from Eq. (10.11), the ideal gas law can be written as

$$PV = nRT. \quad (10.77)$$

Considering all possible differential changes in quantities in Eq. (10.77), we find that

$$P dV + V dP = RT dn + nR dT \quad (10.78)$$

(recall that R is a constant). For constant P and n , Eq. (10.78) implies that $P dV/dT = nR$. Substituting this result into Eq. (10.76) along with $dU = C_V dT$ and the definition of C_P , we arrive at

$$C_P = C_V + nR. \quad (10.79)$$

Equation (10.79) is valid for all situations for which the ideal gas law applies.

Define the parameter γ to be the ratio of specific heats, or

$$\gamma \equiv \frac{C_P}{C_V}. \quad (10.80)$$

For a monatomic gas, we see that $\gamma = 5/3$. If ionization occurs, some of the heat that would normally go into increasing the average kinetic energy of the particles must go into ionizing the atoms instead. Therefore the temperature of the gas, which is a measure of its internal energy, will not rise as rapidly, implying larger values for the specific heats in a partial ionization zone. As both C_P and C_V increase, γ approaches unity.¹⁹

The Adiabatic Gas Law

Since the change in internal energy is independent of the process involved, consider the special case of an **adiabatic process** ($dQ = 0$) for which no heat flows into or out of the mass element. Then the first law of thermodynamics (Eq. 10.73) becomes

$$dU = -P dV.$$

However, from Eq. (10.78) with constant n ,

$$P dV + V dP = nR dT.$$

Also, since $dU = C_V dT$, we have

$$dT = \frac{dU}{C_V} = -\frac{P dV}{C_V}.$$

Combining these results yields

$$P dV + V dP = -\left(\frac{nR}{C_V}\right) P dV,$$

which may be rewritten by using Eqs. (10.79) and (10.80), to give

$$\gamma \frac{dV}{V} = -\frac{dP}{P}. \quad (10.81)$$

Solving this differential equation leads to the adiabatic gas law,

$$PV^\gamma = K, \quad (10.82)$$

where K is a constant. Using the ideal gas law, a second adiabatic relation may be obtained:

$$P = K' T^{\gamma/(\gamma-1)}, \quad (10.83)$$

where K' is another constant. Because of its special role in Eqs. (10.82) and (10.83), γ is often referred to as the “adiabatic gamma,” specifying a particularly simple equation of state.

¹⁹The variation of γ also plays an important role in the dynamic stability of stars. This factor will be discussed further in Section 14.3.

The Adiabatic Sound Speed

Using the results obtained thus far, we can now calculate a sound speed through the material. The sound speed is related to the compressibility of the gas and to its inertia (represented by density) and is given by

$$v_s = \sqrt{B/\rho},$$

where $B \equiv -V(\partial P/\partial V)_{\text{ad}}$ is the bulk modulus of the gas.²⁰ The bulk modulus describes how much the volume of the gas will change with changing pressure. From Eq. (10.81), the adiabatic sound speed becomes

$$v_s = \sqrt{\gamma P/\rho}. \quad (10.84)$$

Example 10.4.2. Assuming a monatomic gas, a typical adiabatic sound speed for the Sun is

$$\bar{v}_s \simeq \left(\frac{5}{3} \frac{\bar{P}}{\bar{\rho}_{\odot}} \right)^{1/2} \simeq 4 \times 10^5 \text{ m s}^{-1},$$

where $\bar{P} \sim P_c/2$ was assumed. The amount of time needed for a sound wave to traverse the radius of the Sun would then be

$$t \simeq R_{\odot}/\bar{v}_s \simeq 29 \text{ minutes}.$$

The Adiabatic Temperature Gradient

Returning now to the specific problem of describing convection, we first consider the situation where a hot convective bubble of gas rises and expands *adiabatically*, meaning that the bubble does not exchange heat with its surroundings. After it has traveled some distance, it finally *thermalizes*, giving up any excess heat as it loses its identity and dissolves into the surrounding gas. Differentiating the ideal gas law (Eq. 10.11) yields an expression involving the bubble's *temperature gradient* (how the bubble's temperature changes with position):

$$\frac{dP}{dr} = -\frac{P}{\mu} \frac{d\mu}{dr} + \frac{P}{\rho} \frac{d\rho}{dr} + \frac{P}{T} \frac{dT}{dr}. \quad (10.85)$$

Using the adiabatic relationship between pressure and density (Eq. 10.82), and recalling that $V \equiv 1/\rho$ is the specific volume, we have

$$P = K\rho^{\gamma}. \quad (10.86)$$

²⁰Formally, the bulk modulus, and therefore the sound speed, must be defined in terms of a process by which pressure varies with volume. Since sound waves typically propagate through a medium too quickly for a significant amount of heat to enter or leave a mass element in the gas, we usually assume that the process is adiabatic.

Differentiating and rewriting, we obtain

$$\frac{dP}{dr} = \gamma \frac{P}{\rho} \frac{d\rho}{dr}. \quad (10.87)$$

If we assume for simplicity that μ is a constant, Eqs. (10.85) and (10.87) may be combined to give *the adiabatic temperature gradient* (designated by the subscript ad)

$$\left. \frac{dT}{dr} \right|_{\text{ad}} = \left(1 - \frac{1}{\gamma} \right) \frac{T}{P} \frac{dP}{dr}. \quad (10.88)$$

Using Eq. (10.6) and the ideal gas law, we finally obtain

$$\boxed{\left. \frac{dT}{dr} \right|_{\text{ad}} = - \left(1 - \frac{1}{\gamma} \right) \frac{\mu m_H}{k} \frac{GM_r}{r^2}.} \quad (10.89)$$

It is sometimes helpful to express Eq. (10.89) in another, equivalent form. Recalling that $g = GM_r/r^2$, $k/\mu m_H = nR$, $\gamma = C_P/C_V$, and $C_P - C_V = nR$, and that n , C_P , and C_V are per unit mass, we have

$$\left. \frac{dT}{dr} \right|_{\text{ad}} = - \frac{g}{C_P}. \quad (10.90)$$

This result describes how the temperature of the gas *inside* the bubble changes as the bubble rises and expands adiabatically.

If the star's *actual temperature gradient* (designated by the subscript act) is *steeper* than the adiabatic temperature gradient given in Eq. (10.89), or

$$\left| \left. \frac{dT}{dr} \right|_{\text{act}} \right| > \left| \left. \frac{dT}{dr} \right|_{\text{ad}} \right|,$$

the temperature gradient is said to be **superadiabatic** (recall that $dT/dr < 0$). It will be shown that in the deep interior of a star, if $|dT/dr|_{\text{act}}$ is just *slightly* larger than $|dT/dr|_{\text{ad}}$, this may be sufficient to carry nearly all of the luminosity by convection. Consequently, it is often the case that either radiation or convection dominates the energy transport in the deep interiors of stars, while the other energy transport mechanism contributes very little to the total energy outflow. The particular mechanism in operation is determined by the temperature gradient. However, near the surface of the star the situation is much more complicated: Both radiation and convection can carry significant amounts of energy simultaneously.

A Criterion for Stellar Convection

Just what condition must be met if convection is to dominate over radiation in the deep interior? When will a hot bubble of gas continue to rise rather than sink back down after being displaced upward? Figure 10.10 shows a convective bubble traveling a distance dr through the surrounding medium. According to Archimedes's principle, if the initial density

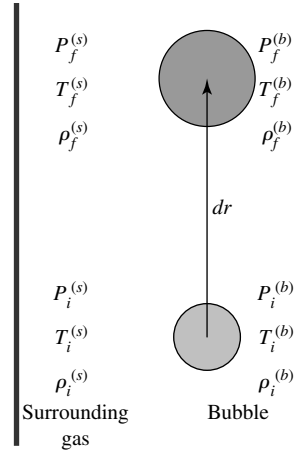


FIGURE 10.10 A convective bubble traveling outward a distance dr . The initial conditions of the bubble are given by $P_i^{(b)}$, $T_i^{(b)}$, and $\rho_i^{(b)}$, for the pressure, temperature, and density, respectively, while the initial conditions for the surrounding gas at the same level are designated by $P_i^{(s)}$, $T_i^{(s)}$, and $\rho_i^{(s)}$, respectively. Final conditions for either the bubble or the surrounding gas are indicated by an f subscript.

of the bubble is less than that of its surroundings ($\rho_i^{(b)} < \rho_i^{(s)}$), it will begin to rise. Now, the buoyant force *per unit volume* exerted on a bubble that is totally submersed in a fluid of density $\rho_i^{(s)}$ is given by

$$f_B = \rho_i^{(s)} g.$$

If we subtract the downward gravitational force per unit volume on the bubble, given by

$$f_g = \rho_i^{(b)} g,$$

the net force per unit volume on the bubble becomes

$$f_{\text{net}} = -g \delta\rho, \quad (10.91)$$

where $\delta\rho \equiv \rho_i^{(b)} - \rho_i^{(s)} < 0$ initially. If, after traveling an infinitesimal distance dr , the bubble now has a greater density than the surrounding material ($\rho_f^{(b)} > \rho_f^{(s)}$), it will sink again and convection will be prohibited. On the other hand, if $\rho_f^{(b)} < \rho_f^{(s)}$, the bubble will continue to rise and convection will result.

To express this condition in terms of temperature gradients, assume that the gas is initially *very nearly in thermal equilibrium*, with $T_i^{(b)} \simeq T_i^{(s)}$ and $\rho_i^{(b)} \simeq \rho_i^{(s)}$. Also assume that the bubble expands adiabatically and that the bubble and surrounding gas pressures are equal at all times, $P_f^{(b)} = P_f^{(s)}$. Now, since it is assumed that the bubble has moved an infinitesimal distance, it is possible to express the final quantities in terms of the initial quantities and

their gradients by using a Taylor expansion. To first order,

$$\rho_f^{(b)} \simeq \rho_i^{(b)} + \left. \frac{d\rho}{dr} \right|^{(b)} dr \quad \text{and} \quad \rho_f^{(s)} \simeq \rho_i^{(s)} + \left. \frac{d\rho}{dr} \right|^{(s)} dr.$$

If the densities inside and outside the bubble remain nearly equal (as is usually the case except near the surfaces of some stars), substituting these results into the convection condition, $\rho_f^{(b)} < \rho_f^{(s)}$, gives

$$\left. \frac{d\rho}{dr} \right|^{(b)} < \left. \frac{d\rho}{dr} \right|^{(s)}. \quad (10.92)$$

We now want to express this solely in terms of quantities for the surroundings. Using Eq. (10.87) for the adiabatically rising bubble to rewrite the left-hand side of (10.92) and using Eq. (10.85) to rewrite the right-hand side (again assuming $d\mu/dr = 0$), we find

$$\frac{1}{\gamma} \frac{\rho_i^{(b)}}{P_i^{(b)}} \left. \frac{dP}{dr} \right|^{(b)} < \frac{\rho_i^{(s)}}{P_i^{(s)}} \left[\left. \frac{dP}{dr} \right|^{(s)} - \frac{P_i^{(s)}}{T_i^{(s)}} \left. \frac{dT}{dr} \right|^{(s)} \right].$$

Recalling that $P^{(b)} = P^{(s)}$ at all times, it is necessary that

$$\left. \frac{dP}{dr} \right|^{(b)} = \left. \frac{dP}{dr} \right|^{(s)} = \frac{dP}{dr},$$

where the superscripts on the pressure gradient have been shown to be redundant. Substituting, and canceling equivalent initial conditions,

$$\frac{1}{\gamma} \frac{dP}{dr} < \frac{dP}{dr} - \frac{P_i^{(s)}}{T_i^{(s)}} \left. \frac{dT}{dr} \right|^{(s)}.$$

Dropping subscripts for initial conditions and superscripts designating the surrounding material, we arrive at the requirement

$$\left(\frac{1}{\gamma} - 1 \right) \frac{dP}{dr} < - \frac{P}{T} \left. \frac{dT}{dr} \right|_{\text{act}}, \quad (10.93)$$

where the temperature gradient is the *actual* temperature gradient of the surrounding gas. Multiplying by the negative quantity $-T/P$ requires that the direction of the inequality be reversed, giving

$$\left(1 - \frac{1}{\gamma} \right) \frac{T}{P} \frac{dP}{dr} > \left. \frac{dT}{dr} \right|_{\text{act}}.$$

But from Eq. (10.88), we see that the left-hand side of the inequality is just the adiabatic temperature gradient. Thus

$$\left. \frac{dT}{dr} \right|_{\text{ad}} > \left. \frac{dT}{dr} \right|_{\text{act}}$$

is the condition for the gas bubble to keep rising. Finally, since $dT/dr < 0$ (the temperature decreases as the stellar radius increases), taking the absolute value of the equation again requires that the direction of the inequality be reversed, or

$$\left| \frac{dT}{dr} \right|_{\text{act}} > \left| \frac{dT}{dr} \right|_{\text{ad}}. \quad (10.94)$$

If the actual temperature gradient is superadiabatic, convection will result, assuming that μ does not vary.

Equation (10.93) may be used to find another useful, and equivalent, condition for convection. Since $dT/dr < 0$ and $1/\gamma - 1 < 0$ (recall that $\gamma > 1$),

$$\frac{T}{P} \left(\frac{dT}{dr} \right)^{-1} \frac{dP}{dr} < -\frac{1}{\gamma^{-1} - 1},$$

which may be simplified to give

$$\frac{T}{P} \frac{dP}{dT} < \frac{\gamma}{\gamma - 1},$$

or, for convection to occur,

$$\frac{d \ln P}{d \ln T} < \frac{\gamma}{\gamma - 1}. \quad (10.95)$$

For an ideal monatomic gas, $\gamma = 5/3$ and convection will occur in some region of a star when $d \ln P / d \ln T < 2.5$. In that case the temperature gradient (dT/dr) is given approximately by Eq. (10.89). When $d \ln P / d \ln T > 2.5$, the region is stable against convection and dT/dr is given by Eq. (10.68).

By comparing Eq. (10.68) for the radiative temperature gradient with either Eq. (10.89) or Eq. (10.90), together with the condition for convection written in terms of the temperature gradient, Eq. (10.94), it is possible to develop some understanding of which conditions are likely to lead to convection over radiation. In general, convection will occur when (1) the stellar opacity is large, implying that an unachievably steep temperature gradient ($|dT/dr|_{\text{act}}$) would be necessary for radiative transport, (2) a region exists where ionization is occurring, causing a large specific heat and a low adiabatic temperature gradient ($|dT/dr|_{\text{ad}}$), and (3) the temperature dependence of the nuclear energy generation rate is large, causing a steep radiative flux gradient and a large temperature gradient. In the atmospheres of many stars, the first two conditions can occur simultaneously, whereas the third condition would occur only deep in stellar interiors. In particular, the third condition can occur when the highly temperature-dependent CNO cycle or triple alpha processes are occurring.

The Mixing-Length Theory of Superadiabatic Convection

It has already been suggested that the temperature gradient must be only slightly superadiabatic in the deep interior in order for convection to carry most of the energy. We will now justify that assertion.

We begin by returning to the fundamental criterion for convection, $\rho_f^{(b)} < \rho_f^{(s)}$. Since the pressure of the bubble and that of its surroundings are always equal, the ideal gas law implies that $T_f^{(b)} > T_f^{(s)}$, assuming thermal equilibrium initially. Therefore, the temperature of the surrounding gas must decrease more rapidly with radius, so

$$\left| \frac{dT}{dr} \right|^{(s)} - \left| \frac{dT}{dr} \right|^{(b)} > 0$$

is required for convection. Since the temperature gradients are negative, we have

$$\frac{dT}{dr} \Big|^{(b)} - \frac{dT}{dr} \Big|^{(s)} > 0.$$

Assuming that the bubble moves essentially adiabatically, and designating the temperature gradient of the surroundings as the actual average temperature gradient of the star, let

$$\frac{dT}{dr} \Big|^{(b)} = \frac{dT}{dr} \Big|_{\text{ad}} \quad \text{and} \quad \frac{dT}{dr} \Big|^{(s)} = \frac{dT}{dr} \Big|_{\text{act}}.$$

After the bubble travels a distance dr , its temperature will exceed the temperature of the surrounding gas by²¹

$$\delta T = \left(\frac{dT}{dr} \Big|_{\text{ad}} - \frac{dT}{dr} \Big|_{\text{act}} \right) dr = \delta \left(\frac{dT}{dr} \right) dr. \quad (10.96)$$

We use δ here to indicate the difference between the value of a quantity associated with the bubble and the same quantity associated with the surroundings, both determined at a specified radius r , just as was done for Eq. (10.91).

Now assume that a hot, rising bubble travels some distance

$$\ell = \alpha H_P$$

before dissipating, at which point it thermalizes with its surroundings, giving up its excess heat at constant pressure (since $P^{(b)} = P^{(s)}$ at all times). The distance ℓ is called the **mixing length**, H_P is the pressure scale height (see Eq. 10.70), and

$$\alpha \equiv \ell / H_P,$$

the ratio of mixing length to pressure scale height is an adjustable parameter, or *free parameter*, generally assumed to be of order unity. (From comparisons of numerical stellar models with observations, values of $0.5 < \alpha < 3$ are typical.)

After the bubble travels one mixing length, the excess heat flow *per unit volume* from the bubble into its surroundings is just

$$\delta q = (C_P \delta T) \rho,$$

²¹ In some texts, $\delta \left(\frac{dT}{dr} \right) \equiv \Delta \nabla T$.

where δT is calculated from Eq. (10.96) by substituting ℓ for dr . Multiplying by the average velocity $\bar{v}_c$ of the convective bubble, we obtain the convective flux (the amount of energy per unit area per unit time carried by a bubble):

$$F_c = \delta q \bar{v}_c = (C_P \delta T) \rho \bar{v}_c. \quad (10.97)$$

Note that $\rho \bar{v}$ is a *mass flux*, or the amount of mass per second that crosses a unit area oriented perpendicular to the direction of the flow. Mass flux is a quantity that is often encountered in fluid mechanics.

The average velocity $\bar{v}$ may be found from the net force *per unit volume*, f_{net} , acting on the bubble. Using the ideal gas law and assuming constant μ , we can write

$$\delta P = \frac{P}{\rho} \delta \rho + \frac{P}{T} \delta T.$$

Since the pressure is always equal between the bubble and its surroundings, $\delta P \equiv P^{(b)} - P^{(s)} = 0$. Thus

$$\delta \rho = -\frac{\rho}{T} \delta T.$$

From Eq. (10.91),

$$f_{\text{net}} = \frac{\rho g}{T} \delta T.$$

However, we assumed that the initial temperature difference between the bubble and its surroundings is essentially zero, or $\delta T_i \approx 0$. Consequently the buoyant force must also be quite close to zero initially. Since f_{net} increases linearly with δT , we may take an average over the distance ℓ between the initial and final positions, or

$$\langle f_{\text{net}} \rangle = \frac{1}{2} \frac{\rho g}{T} \delta T_f.$$

Neglecting viscous forces, the work done per unit volume by the buoyant force over the distance ℓ goes into the kinetic energy of the bubble, or

$$\frac{1}{2} \rho v_f^2 = \langle f_{\text{net}} \rangle \ell.$$

Choosing an *average* kinetic energy over one mixing length leads to some average value of v^2 , namely βv^2 , where β has a value in the range $0 < \beta < 1$. Now the average convective bubble velocity becomes

$$\bar{v}_c = \left(\frac{2\beta \langle f_{\text{net}} \rangle \ell}{\rho} \right)^{1/2}.$$

Substituting the net force per unit volume, using Eq. (10.96) with $dr = \ell$, and rearranging,

we have

$$\begin{aligned}\bar{v}_c &= \left(\frac{\beta g}{T}\right)^{1/2} \left[\delta \left(\frac{dT}{dr}\right)\right]^{1/2} \ell \\ &= \beta^{1/2} \left(\frac{T}{g}\right)^{1/2} \left(\frac{k}{\mu m_H}\right) \left[\delta \left(\frac{dT}{dr}\right)\right]^{1/2} \alpha,\end{aligned}\quad (10.98)$$

where we obtained the last equation by replacing the mixing length with αH_P and using Eq. (10.70) together with the ideal gas law.

After some manipulation, Eqs. (10.97) and (10.98) finally yield an expression for the convective flux:

$$F_c = \rho C_P \left(\frac{k}{\mu m_H}\right)^2 \left(\frac{T}{g}\right)^{3/2} \beta^{1/2} \left[\delta \left(\frac{dT}{dr}\right)\right]^{3/2} \alpha^2. \quad (10.99)$$

Fortunately, F_c is not very sensitive to β , but it does depend strongly on α and $\delta(dT/dr)$.

The derivation leading to the prescription for the convective flux given by Eq. (10.99) is known as the **mixing-length theory**. Although basically a phenomenological theory containing arbitrary constants, the mixing-length theory is generally quite successful in predicting the results of observations.

To evaluate F_c , we still need to know the difference between the temperature gradients of the bubble and its surroundings. Suppose, for simplicity, that *all* of the flux is carried by convection, so that

$$F_c = \frac{L_r}{4\pi r^2},$$

where L_r is the interior luminosity. This will allow us to estimate the difference in temperature gradients needed for this special case. Solving Eq. (10.99) for the temperature gradient difference gives

$$\delta \left(\frac{dT}{dr}\right) = \left[\frac{L_r}{4\pi r^2} \frac{1}{\rho C_P \alpha^2} \left(\frac{\mu m_H}{k}\right)^2 \left(\frac{g}{T}\right)^{3/2} \beta^{-1/2} \right]^{2/3}. \quad (10.100)$$

Dividing Eq. (10.100) by Eq. (10.90) for the adiabatic temperature gradient gives an estimate of how superadiabatic the actual temperature gradient must be to carry all of the flux by convection alone:

$$\frac{\delta(dT/dr)}{|dT/dr|_{\text{ad}}} = \left(\frac{L_r}{4\pi r^2}\right)^{2/3} C_P^{1/3} \rho^{-2/3} \alpha^{-4/3} \left(\frac{\mu m_H}{k}\right)^{4/3} \frac{1}{T} \beta^{-1/3}.$$

Example 10.4.3. Using values typical of the base of the Sun's convection zone, assuming a monatomic gas throughout, and assuming $\alpha = 1$ and $\beta = 1/2$, we can estimate a characteristic adiabatic temperature gradient, the degree to which the actual gradient is superadiabatic, and the convective bubble velocity.

Assume that $M_r = 0.976 M_\odot$, $L_r = 1 L_\odot$, $r = 0.714 R_\odot$, $g = GM_r/r^2 = 525 \text{ m s}^{-2}$, $C_P = 5nR/2$, $P = 5.59 \times 10^{12} \text{ N m}^{-2}$, $\rho = 187 \text{ kg m}^{-3}$, $\mu = 0.606$, and $T = 2.18 \times 10^6 \text{ K}$. Then, from Eq. (10.90),

$$\left| \frac{dT}{dr} \right|_{\text{ad}} \sim 0.015 \text{ K m}^{-1},$$

and from Eq. (10.100),

$$\delta \left(\frac{dT}{dr} \right) \sim 6.7 \times 10^{-9} \text{ K m}^{-1}.$$

The relative amount by which the actual temperature gradient is superadiabatic is then

$$\frac{\delta(dT/dr)}{|dT/dr|_{\text{ad}}} \sim 4.4 \times 10^{-7}.$$

For parameters appropriate for the deep interior, convection is certainly adequately approximated by the adiabatic temperature gradient.

The convective velocity needed to carry all of the convective flux is found from Eq. (10.98),

$$\bar{v}_c \sim 50 \text{ m s}^{-1} \sim 10^{-4} v_s,$$

where v_s is the local solar sound speed (see Eq. 10.84).

Near the surface of a star, where the presence of ionization results in a larger value for C_P and where ρ and T get much smaller, the ratio of the superadiabatic excess to the adiabatic gradient can become significantly larger, with the convective velocity possibly approaching the sound speed. In this situation, a detailed study of the relative amounts of convective and radiative flux must be considered. This will not be discussed further here.

Although the mixing length theory is adequate for many problems, it is incomplete. For instance, α and β are *free parameters* that must be chosen for a particular problem; they may even vary throughout the star. There are also stellar conditions for which the time-independent mixing length theory is inherently unsatisfactory. As one example, consider stellar pulsations; during a star's pulsation cycle the outer layers of the star are oscillating with periods comparable to the timescale for convection, given by $t_c = \ell/\bar{v}_c$. In such cases, rapid changes in the physical conditions in the star directly couple to the driving of the convective bubbles, which in turn alters the structure of the star. Although much effort (and some progress) has been made in developing a full, time-dependent convection theory for stellar interiors, at present no theory exists that completely describes this complex behavior. Much work remains to be done in understanding the important details of stellar convection.

10.5 ■ STELLAR MODEL BUILDING

We have now derived all of the fundamental differential equations of stellar structure. These equations, together with a set of relations describing the physical properties of the stellar material, may be solved to obtain a theoretical stellar model.

A Summary of the Equations of Stellar Structure

For convenience, the basic *time-independent* (static) stellar structure equations are summarized:

$$\frac{dP}{dr} = -G \frac{M_r \rho}{r^2} \quad (10.6)$$

$$\frac{dM_r}{dr} = 4\pi r^2 \rho \quad (10.7)$$

$$\frac{dL_r}{dr} = 4\pi r^2 \rho \epsilon \quad (10.36)$$

$$\frac{dT}{dr} = -\frac{3}{4ac} \frac{\bar{\kappa} \rho}{T^3} \frac{L_r}{4\pi r^2} \quad (\text{radiation}) \quad (10.68)$$

$$= -\left(1 - \frac{1}{\gamma}\right) \frac{\mu m_H}{k} \frac{GM_r}{r^2} \quad (\text{adiabatic convection}) \quad (10.89)$$

The last equation assumes that the convective temperature gradient is purely adiabatic and is applied when

$$\frac{d \ln P}{d \ln T} < \frac{\gamma}{\gamma - 1}. \quad (10.95)$$

If the star is static, as assumed above, then $\epsilon = \epsilon_{\text{nuclear}}$. However, if the structure of the stellar model is changing over time, we must include the energy contribution due to gravity, $\epsilon = \epsilon_{\text{nuclear}} + \epsilon_{\text{gravity}}$. The introduction of the gravitational energy term adds an explicit time dependence to the equations that is not present in the purely static case. This can be seen by realizing that the virial theorem requires that one-half of the gravitational potential energy that is lost must be converted into heat. The *rate* of energy production (per unit mass) by gravity is then dQ/dt . Therefore $\epsilon_{\text{gravity}} = -dQ/dt$, the minus sign indicating that heat is liberated *from* the material.

Entropy

As a note of interest, it is often useful to express the gravitational energy generation rate in terms of the change in the **entropy** per unit mass (the specific entropy), defined by²²

$$dS \equiv \frac{dQ}{T}. \quad (10.101)$$

Then the energy generation rate is seen to be due to the change in entropy of the material, or

$$\epsilon_{\text{gravity}} = -T \frac{dS}{dt}. \quad (10.102)$$

²²Although dQ is an inexact differential, it can be shown that the entropy is a state function.